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Face Recognition in Low-Light Images: Enhancing Visibility Using Deep Learning-Based Image Enhancement

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Declaration Form

University of Dhaka

We, the undersigned, hereby declare that the work presented here is performed under the supervision of our project supervisor. We would also like to state the fact that no part of this work has been submitted in the awarding of any other degrees or diploma, neither has it been published elsewhere.

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Abstract

Face recognition in low-light conditions poses significant challenges due to reduced visibility and noise, impacting the performance of conventional recognition systems. This thesis proposes a discriminative multi-level face loss function that preserves facial identity information during enhancement. Building on the HVI-CIDNet architecture [47], we integrate multi-level feature reconstruction, supervised contrastive learning, and triplet loss to maintain discriminative face features. We introduce a physics-based low-light synthesis pipeline generating multi-level difficulty datasets (easy/medium/hard) with realistic Poisson-Gaussian noise, white balance failure, and blur degradation . Extensive experiments on LFW face verification demonstrate that our face_loss5 model achieves EER of 0.25% (medium), 0.55% (hard), and 0.35% (mixed), significantly outperforming the baseline while maintaining competitive image quality. This work seeks to contribute to the field of computer vision by advancing reliable face recognition in adverse environments, with applications in security, surveillance, and human-computer interaction.

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Finally, we express our gratitude to the Almighty for offering us strength and directions to overcome all the difficulties.

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Abbreviations

LLIE	**L**ow **L**ight **I**mage **E**nhancement
FR	**F**ace **R**ecognition
HE	**H**istogram **E**qualization
ZeroDCE	**Zero** Reference **D**eep **C**urve **E**stimation
GSAD	**G**lobal **S**tructure-**A**ware **D**iffusion
PSNR	**P**eak **S**ignal to **N**oise **R**atio
SSIM	**S**tructural **S**imilarity **I**ndex **M**easure
LPIPS	**L**earned **P**erceptual **Image** **P**atch **S**imilarity
HVI	**H**orizontal/**V**ertical **I**ntensity
CIDNet	**C**olor and **I**ntensity **D**ecoupling **Net**work
LCA	**L**ighten **C**ross **A**ttention
PAB	**P**rior-guided **A**ttention **B**lock
CNN	**C**onvolutional **N**eural **N**etwork
ResNet	**Res**idual **Net**work
LFW	**L**abelled **F**ace in the **W**ild

Chapter 1

Introduction

Facial recognition is an important area of research in Computer Vision and Artificial Intelligence, with applications in security, surveillance, authentication and human - computer interaction. This system usually recognizes or identifies a person based on their facial features by comparing an input image with a database of stored images. While recent advancements in deep - learning based approaches such as CNNs and margin based loss functions (For example, ArcFace, CosFace) shows remarkable performance and accuracy in controlled environment with good lighting conditions, their performance significantly degrades in low - light conditions due to loss of texture details, low contrast and increased noise in the images. Thus it becomes difficult for the facial recognition models to extract discriminative features accurately. Since many real - life applications work under unconstrained environments, such as nighttime surveillance, CCTV footage, low - lit indoor space, poor weather conditions, facial recognition in low - light environments remains an active and challenging research problem.

Low - Light Image Enhancement (LLIE) works as a preprocessing step in improving the accuracy of the facial recognition models in Low- Light conditions. It improves image visibility, enhances contrast and recovers facial details before feeding the image into the facial recognition model. Early approaches to LLIE relied on **Histogram Equalization (HE)** and **Retinex theory** [26] which increase brightness and contrast but also introduce problems such as artifacts, overenhancement, noise amplification and unnatural color. Although these methods are computationally efficient, they fail to generalize across various illumination conditions. With the advancement of deep learning, LLIE research goes through a paradigm shift towards learning based models that require large datasets to learn illumination correction. **Deep Retinex - Net (Wei et al., 2018)** [45] was one of the first Deep Learning models based on Retinex Decomposition to predict illumination and reflectance maps. Following this, **EnlightenGAN (Jiang et al., 2021)**[20] introduced an unsupervised GAN-based approach that can enhance illumination without requiring a labeled dataset to learn the model. Another breakthrough was **ZeroDCE (Zero Reference Deep Curve Estimation)** [12] which also eliminates the need for reference images by predicting pixel-wise curve parameters in an unsupervised manner. These state-of-the-art methods significantly outperform the traditional approaches and techniques in terms of both visual quality and performance metrics such as PSNR and SSIM. However, integrating LLIE in the facial recognition pipeline introduces new challenges. Enhancement approaches must preserve specific facial features crucial for recognition, at the same time avoiding distortion that can lead to mismatch. Many researchers have explored both the independent enhancement models and joint optimization of enhancement and recognition, but achieving an optimal balance between the visual quality and recognition accuracy still remains a challenging task.

To address this challenge, we propose a face-recognition-aware low-light enhancement framework that integrates an advanced LLIE model **HVI-CIDNet** with a frozen state-of-the-art deep face recognition network **AdaFace**. Although the conventional approaches perform the two - stage pipeline independently, our approach explicitly supervises the enhancement process using multi-level facial embeddings extracted from the face recognition model. This allows the enhancement network to preserve identity-discriminative features while improving visual quality. Additionally, we train and evaluate the framework using a physics-based low-light synthesis pipeline that generates realistic low-light facial images with controllable degradation factors such as illumination reduction, Poisson-Gaussian noise, white balance failure, and blur.

1.1 Motivations

In various real-world applications such as security surveillance, law enforcement, identity verification, etc. facial recognition is an essential technology. Although modern facial recognition models like Adaface, ArcFace, InsightFace, Facenet achieve near-perfect accuracy on various metrics, their performance degrades sharply in low-light environments due to insufficient illumination, noise, and distortion of facial features.

There are two challenges in low-light conditions. Firstly, the visual quality of the images or video frames is very poor, making it difficult for even humans to recognize faces. Secondly, all the popular facial recognition models are trained on well-lit datasets, so they cannot extract meaningful facial features from low-light inputs. Recent LLIE models provide significant improvement in visual quality, but most are designed for image-level enhancement without specific consideration for facial structures or recognition performance.

So, the motivation for integrating LLIE with facial recognition comes from the need for creating a system capable of :

- Enhancing Low- Light images while preserving facial features crucial for facial recognition
- Suppressing noise without sacrificing facial integrity, important for reliable recognition in challenging environment
- Creating a dataset specifically for the framework and training on the dataset made for Low - light facial recognition
- Providing a practical framework for real - world applications such as security surveillance, autonomous attendance system and so on.

By combining advanced LLIE techniques with facial recognition, we aim to reduce the performance gap between ideal and low-light conditions making facial recognition more reliable, efficient, and robust in challenging real - world scenarios.

1.2 Problem Formulation

Facial recognition from low-light images presents a compound problem involving both image quality degradation and facial feature extraction reliability. Formally, given a set of facial images or video frames,

$$\mathcal{I}_{low} = \{I_1, I_2, \ldots, I_n\}$$

captured under low-light conditions, the goal is to obtain a set of facial embeddings

$$\{E_1, E_2, \ldots, E_n\}$$

corresponding to each input sample, such that the embeddings can be matched against a pre-existing gallery database of known identities.

So the problem can be formulated as a two step recognition aware enhancement task :

1. **Enhancement:** Learn a mapping $H : I_i \to \hat{I}_i$, where $\hat{I}_i$ represents the enhanced image with improved visual quality that also preserves identity-related facial features.
2. **Recognition:** Extract embedding $E_i = R(\hat{I}_i)$ using a facial recognition model R, where $E_i \in \mathbb{R}^d$ represents the identity embedding used for matching.

Instead of optimizing the enhancement and recognition stages independently, the objective of this work is to supervise the enhancement process using the recognition model. In a broader sense, the enhancement function H is optimized not only to minimize image-level reconstruction error, but also to keep discriminative structure in the embedding space of R. The overall goal is to improve recognition accuracy under low-light conditions while minimizing visual artifacts and feature space distortion.

1.3 Objectives

The main objective of this project is to create a system that can enhance images taken in low-light conditions, making sure that important facial features are preserved for accurate face recognition.

The particular objectives are as follows -

1. **Create a Recognition-Aware Low-Light Enhancement Module:** Choose an advanced LLIE model and incorporate face-level supervision to adapt it to maintain facial level features more efficiently .

2. **Combine Enhancement with Face Embedding Supervision:** To guide the enhancement process, a state of the art face recognition model is selected to integrate with the enhancement process.

3. **Create and Utilize a Realistic Low-Light Dataset:** Create a physics-based low - light face dataset with controlled degradation and noise factors for training and evaluation.

4. **Evaluate the Result of Recognition - Aware Enhancement :** To compare the recognition performance of low - light facial images before and after enhancement, use verification metrics like EER and TAR to quantitatively .

By achieving the objectives, we hope to bridge the gap between low - light image enhancement and face recognition with identity preservation , creating a reliable and robust solution for real-world systems operating under low-light conditions .

1.4 Contributions

Low Light Image Enhancement and Facial Recognition both lie in the domain of Computer Vision of Artificial Intelligence. Instead of treating these as two independent tasks, we propose a combined recognition-aware enhancement framework that efficiently preserves identity information during the enhancement process.

The main contributions can be listed as :

1. Development of a novel face-recognition-aware low-light enhancement framework. We design a system that integrates a state-of-the-art LLIE model with a deep face recognition network and jointly optimizes enhancement using identity-based supervision.

2. Proposal of a discriminative multi-level face loss. The loss function combines feature reconstruction, supervised contrastive learning, and triplet margin loss to preserve discriminative facial embeddings during enhancement.

3. Introduction of a physics-based low-light synthesis pipeline. We generate realistic low-light facial images using illumination reduction, Poisson-Gaussian noise, white balance failure, and blur degradation with controllable difficulty levels.

4. Extensive experimental evaluation on low-light face verification. We demonstrate that the proposed method significantly improves recognition accuracy compared to conventional enhancement approaches while maintaining competitive image quality.

1.5 Challenges

The main challenges addressed in this thesis can be stated as :

1. **Illumination-Induced Degradation:** Low-light images suffer from reduced contrast, color distortion, and high sensor noise, making it difficult to extract reliable and discriminative facial features.

2. **Feature Space Compression:** Conventional enhancement losses optimized for perceptual quality tend to compress the embedding space of face representations, reducing inter-class separability.

3. **Identity Preservation:** Enhanced images must preserve subtle identity-related facial cues rather than only improving visual appearance.

4. **Dataset Realism:** Most publicly available low-light datasets lack realistic sensor noise and identity labels, making it difficult to train and evaluate recognition-aware enhancement models.

1.6 Organization

The thesis is organized in a systematic way. The organization can be seen as:

Chapter 1 introduces low-Light image enhancement for facial recognition, providing motivation, problem formulation, objectives, contributions, and report organization.

Chapter 2 is the literature review section that discusses prior works in LLIE, video-based enhancement, and facial recognition under challenging lighting conditions. It reviews classical methods such as histogram equalization and Retinex theory, deep learning-based LLIE approaches, and state-of-the-art video-level enhancement techniques.

Chapter 3 explains in detail the pipeline including video processing, low-light image enhancement, facial recognition based framework, embedding techniques and integration strategy.

Chapter 4 provides the implementation strategy used in detail for both the pipeline and creation of the database. It also includes the current limitations and future plans for the project.

Chapter 5 Presents dataset details, evaluation metrics, and performance analysis of enhancement and recognition, including quantitative results and qualitative visual examples.

Chapter 6 summarizes key findings, contributions, and suggests potential improvements and directions for future research

Chapter 2

Literature Review

Face recognition (FR) under low-light conditions faces challenges rising from severe image degradation . These include diminished contrast, amplified sensor noise, color distortion, and the loss of identity-related facial details, all of which significantly reduce recognition accuracy . Low - light image enhancement (LLIE) methods are commonly applied to improve visual quality in such scenarios. However, most existing LLIE approaches are optimized primarily for perceptual quality and do not explicitly consider the requirements of downstream face recognition tasks. As a result, enhanced images may appear visually pleasing while still degrading the discriminative structure of face embeddings .

The integration of LLIE with FR systems , particularly from the perspective of identity preservation and embedding space geometry , remains an underexplored research area . This chapter provides a comprehensive review of related literature , organized into sections covering LLIE techniques, general deep face recognition methods , recognition aware enhancement strategies, foundational background concepts, and an analysis of existing limitations . The emphasis is on recent deep learning developments , including models that incorporate attention mechanisms and representation learning frameworks , which aim to preserve semantic and identity - related features while performing enhancement.

2.1 Low-Light Image Enhancement

LLIE aims to recover clarity and realism in images taken in low - light environments . Early DL approaches, such as LLNet [30] , used autoencoders for low - light enhancement and denoising , paving the way for models like RetinexNet Conventional techniques, such as histogram equalization or those rooted in Retinex theory [26] , break down images into reflectance and illumination layers . Jobson et al. [21] extended Retinex with a center surround approach , improving robustness but still relying on fixed priors , limiting adaptability in complex scenes. However, these often depend on manually designed assumptions that restrict their flexibility in varied real-world settings. Non-DL methods like LIME [14] estimate illumination maps to enhance low - light images but struggle with over enhancement, as noted in comparisons with Retinex - Net.

2.1.1 Retinexformer

Cai et al. introduced Retinexformer to resolve issues in earlier Retinex-based deep models, which often ignore degradations in shadowy areas or require multiple processing stages, and rely on convolutions that miss global interactions [1]. The framework is based on a one-stage Retinex approach that estimates illumination to direct both the enhancement and restoration of corrupted elements.

Central to the design is the Illumination-Guided Transformer (IGT), which employs the predicted illumination to steer non-local interactions through self-attention mechanisms. Self-attention calculates importance weights across all input elements, enabling the capture of widespread dependencies that are crucial for handling uneven lighting. This allows the model to focus on regions with varying illumination levels effectively.

2.1.2 Global Structure-Aware Diffusion Process (GSAD)

Hou et al. proposed this diffusion-based model to enhance low - light images by boosting quality , suppressing noise, and increasing contrast through regularization of ordinary differential equation (ODE) paths [18] . The concept involves using low curvature ODE trajectories for stable diffusion processes , incorporating global structure-aware terms that consider non-local image elements to maintain details and heighten contrast . An uncertainty guided adjustment eases constraints in highly degraded areas for better flexibility.

Benefits include reliable trajectory stability and detail retention. However, the iterative diffusion steps demand significant computation, limiting suitability for real-time video tasks.

The figure 2.1 shows the structure of GSAD, depicting the regularization of ODE trajectories with global structure-aware terms and uncertainty guidance [18].

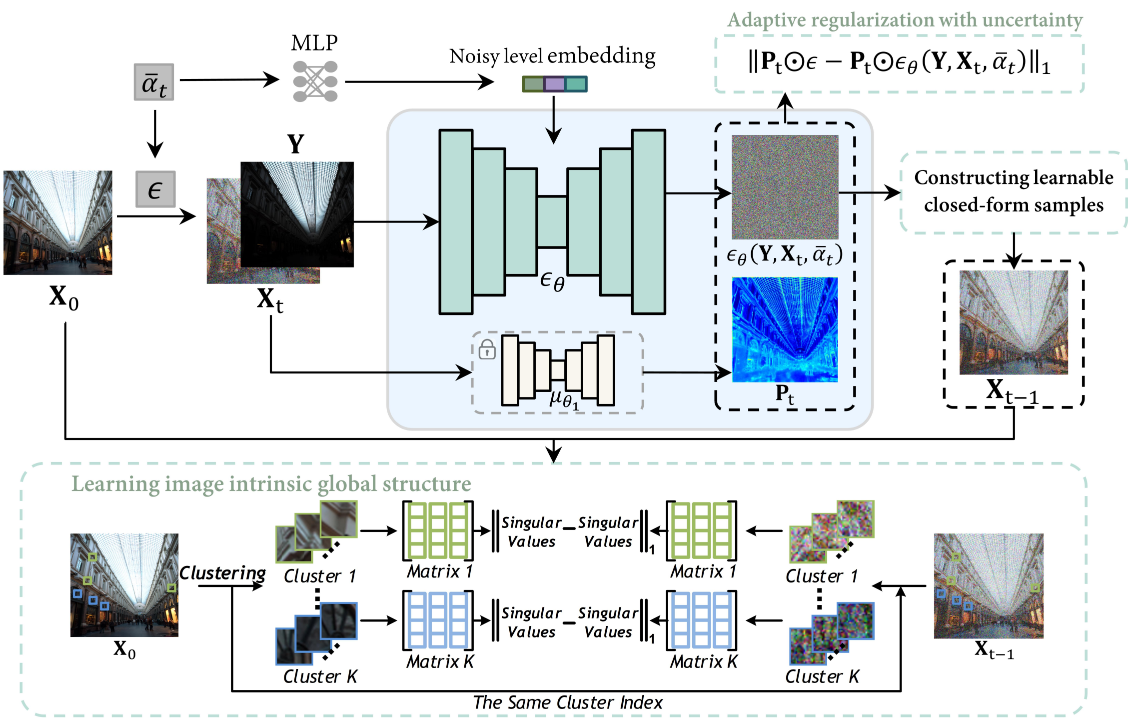


FIGURE 2.1: Global Structure-Aware Diffusion Process architecture.

2.1.3 HVI-CIDNet

Yan et al. created HVI-CIDNet to combat color distortions in standard spaces like sRGB (which can cause biases and brightness artifacts) and HSV (prone to red/black noise) during LLIE [47]. The innovation lies in the Horizontal/Vertical-Intensity (HVI) color space, designed to separate chromatic and intensity data more effectively. Polarized hue-saturation maps enforce minimal distances for red coordinates to eliminate red artifacts, while a learnable intensity component compresses dark regions to avoid black artifacts.

2.1.3.1 RGB to HVI and HVI to RGB Conversion Algorithms

This section presents the algorithms and equations for converting images from the RGB (sRGB) color space to the Horizontal/Vertical-Intensity (HVI) color space and vice versa, as used in low-light image enhancement tasks.

The algorithms 1 and 2 contain the steps to follow for conversion from RGB color space to HVI and HVI to RGB respectively.

RGB to HVI conversion:

Algorithm 1: RGB to HVI Transformation

`RGB2HVI(`I_{RGB}`)`

Input : I_{RGB}: Image in RGB space

Output: I_{HVI}: Image in HVI space

$I_{\max}(x) \leftarrow \max\{R(x), G(x), B(x)\}$

$S(x) \leftarrow \frac{\Delta}{I_{\max}}, \quad \Delta = I_{\max} - \min\{R, G, B\}$

$$H(x) \leftarrow \begin{cases} 0, & S = 0 \\ \frac{G-B}{\Delta} \bmod 6, & I_{\max} = R \\ 2 + \frac{B-R}{\Delta}, & I_{\max} = G \\ 4 + \frac{R-G}{\Delta}, & I_{\max} = B \end{cases}$$

$h \leftarrow \cos(\frac{\pi H}{3}), \quad v \leftarrow \sin(\frac{\pi H}{3})$

$C_k(x) \leftarrow \sqrt[k]{\sin(\frac{\pi I_{\max}(x)}{2}) + \varepsilon}$

$\hat{H} \leftarrow C_k \odot S \odot h, \quad \hat{V} \leftarrow C_k \odot S \odot v$

$I_{HVI} \leftarrow [\hat{H}, \hat{V}, I_{\max}]$

return I_{HVI}

HVI to RGB conversion:

Algorithm 2: HVI to RGB Transformation (Perceptual-inverse)

`HVI2RGB(`I_{HVI}`)`

Input : I_{HVI} : Image in HVI space

Output: I_{RGB} : Image in RGB space

$\hat{h} \leftarrow \frac{\hat{H}}{C_k+\varepsilon}, \quad \hat{v} \leftarrow \frac{\hat{V}}{C_k+\varepsilon}$

$H \leftarrow \arctan(\frac{\hat{v}}{\hat{h}}) \bmod 1, \quad S \leftarrow \alpha_S\sqrt{\hat{h}^2 + \hat{v}^2}, \quad V \leftarrow \alpha_I I_{\max}$

Convert (H, S, V) to RGB using HSV $\rightarrow$ RGB mapping

return I_{RGB}

The Color and Intensity Decoupling Network (CIDNet) learns mappings in this space. It decouples the color space into HV channel and I channel, then uses specialized sub-networks: one for chromatic restoration and one for intensity tuning. Both employ a UNet structure with parallel channels for HV and I, fused via Lighten Cross Attention (LCA) blocks that enhance local contrast and details through cross-attention, which aligns features from different sources by computing query-key-value interactions.

The figure 2.2 illustrates the decoupling mechanism, parallel sub-networks for HV and I channels used by HVI-CIDNet. [47] and figure 2.3 dives deep into the Cross Attention-based Lighten Cross Attention (LCA) block used in the Enhancement Network for feature integration.

Across 10 datasets including LOL and SICE, it outperformed competitors, achieving PSNR of 23.81 and SSIM of 0.912 on LOL eval15, demonstrating excellent noise reduction and structural fidelity. Strengths encompass effective artifact removal and broad applicability. A limitation is its orientation toward static images, without built-in support for temporal aspects in videos. KinD [51], a Retinex-inspired DL method, focuses on noise suppression and detail preservation, serving as a benchmark for HVI-CIDNet.

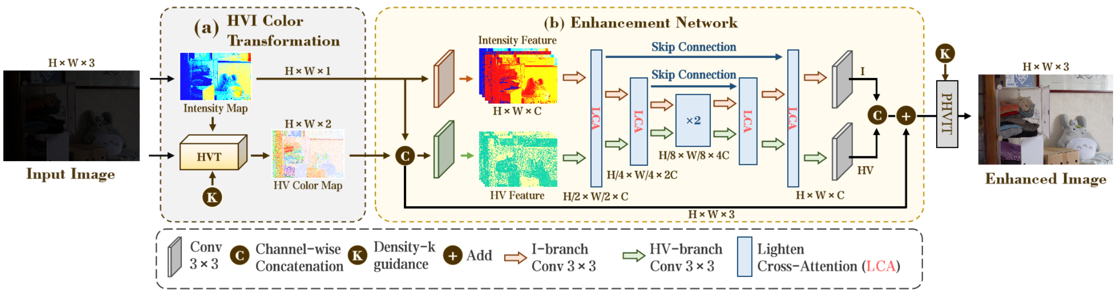


FIGURE 2.2: HVI-CIDNet architecture.

2.1.4 HVI-CIDNet+

This extension by Yan et al. targets even darker conditions, adding a Prior-guided Attention Block (PAB) that uses pre-trained vision-language models for contextual and degradation-specific guidance via cross-attention, and a Region Refinement

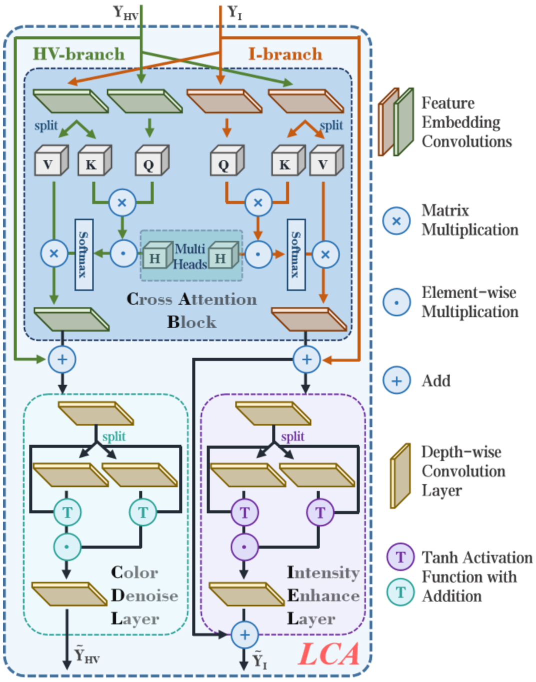


FIGURE 2.3: HVI-CIDNet architecture.

Block that blends convolutions for detail-rich zones with self-attention for sparse ones [48]. Evaluations on 10 datasets confirmed superior results. Advantages include advanced handling of extreme low light. As a preprint awaiting publication and post-dating our research start, we rely on the established HVI-CIDNet from CVPR 2025 for its proven track record.

The figure 2.4 shows the internal archicture of HVI-CIDNet+ and 2.5 shows the Prior-guided Attention Block (PAB) structure in detail.

Justification for HVI-CIDNet HVI-CIDNet stands out for our use due to its excellence in severe low light, lightweight build for per-frame video application, and FR integration potential. It edges out Retinexformer (over-dependent on illumination) and diffusion models (resource-heavy). Yet, its static focus necessitates video adaptations for consistency, which our approach provides.

These LLIE progresses integrate transformer features like self-attention for comprehensive feature grasping and cross-attention for multi-level merging, as in Vision Transformers (ViT) [7], advancing noise and color accuracy. Still, most are image-focused, highlighting the demand for video-oriented systems.

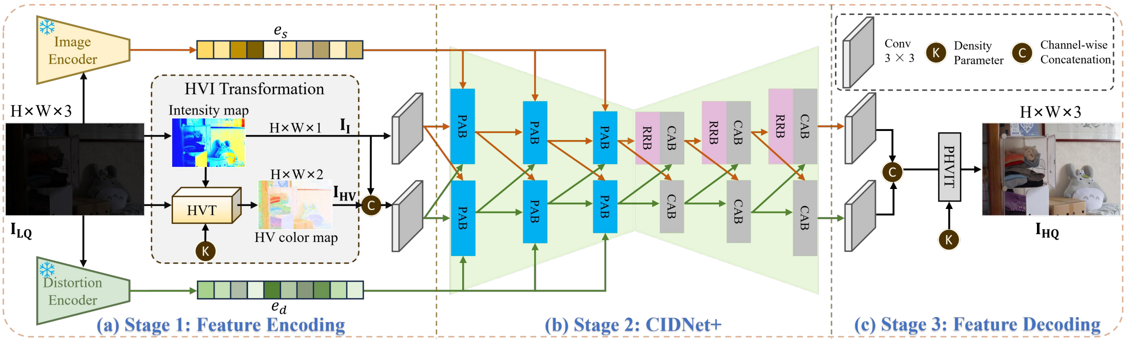


FIGURE 2.4: HVI-CIDNet+ extensions for enhanced processing in extreme darkness.

2.2 Face Recognition Techniques

FR has progressed from classic methods to deep learning frameworks, with convolutional neural networks (CNNs) leading in feature extraction. Early DL face recognition, such as DeepFace [38], used CNNs to achieve near-human performance, setting the stage for architectures like ResNet [17]. ResNet, by He et al., introduces residual learning to train deeper networks by incorporating shortcut connections that facilitate optimization and avert gradient vanishing [17]. In residual blocks, layers learn differences relative to the input (F(x) + x), allowing networks up to 152 layers deep without performance drops. This yields low error rates on ImageNet and boosts tasks like detection. For FR, ResNet extracts layered features, from low-level edges to high-level semantic traits, enabling robust identity discrimination. FaceNet [36] introduced triplet loss for embedding extraction, a technique adopted in InsightFace [13] for robust identity discrimination.

2.2.1 AdaFace

AdaFace, proposed by Kim et al., is a state-of-the-art deep face recognition framework designed to address the limitations of fixed margin softmax loss under unconstrained and low-quality imaging conditions [24] . Conventional margin based methods such as ArcFace and CosFace enforce a constant angular margin for all training samples , assuming uniform image quality across the dataset. However, in real world scenarios, facial images often shows significant variations in quality

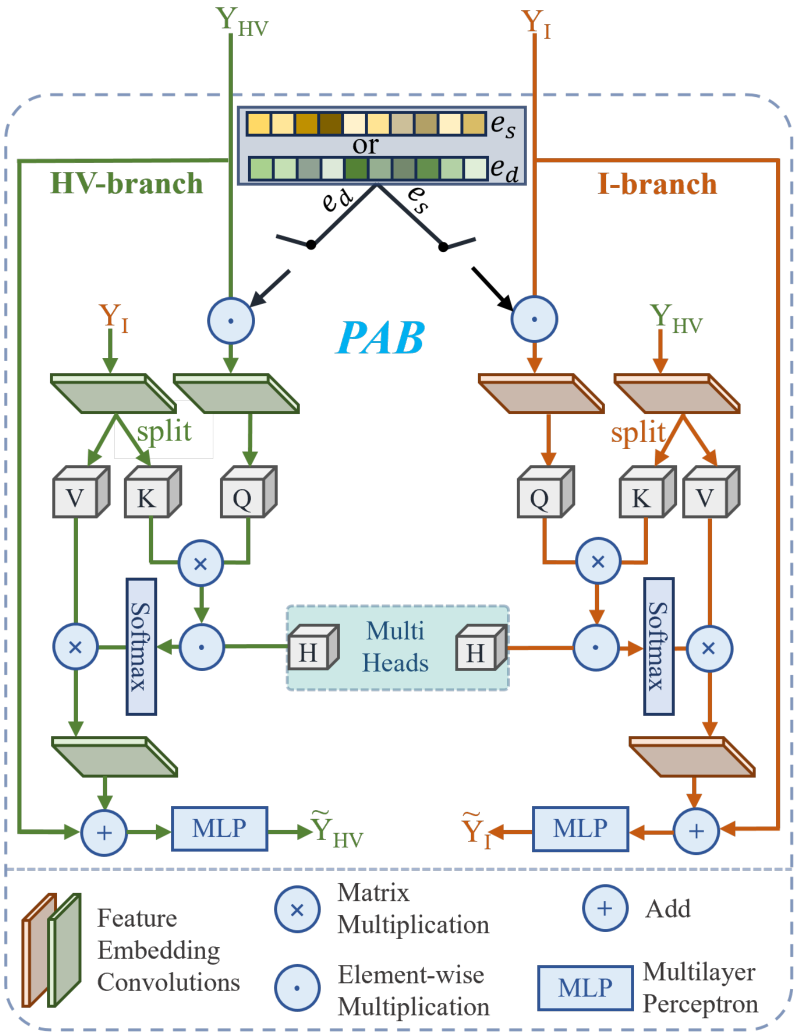


FIGURE 2.5: HVI-CIDNet+ Prior-guided Attention Block and Region Refinement Block

due to factors such as illumination changes, motion blur, pose variation, and sensor noise . Applying a uniform margin to such heterogeneous data leads to unstable optimization and degraded generalization performance .

The core innovation of AdaFace lies in its *quality adaptive margin learning* mechanism , which dynamically adjusts the classification margin for each sample based on its estimated feature quality . Instead of relying on external quality assessment models , AdaFace uses the L2 norm of the embedding vector as a proxy for sample quality , motivated by the observation that high quality facial images tend to produce embeddings with larger magnitudes , while degraded images yield embeddings with smaller norms.

Formally, given a facial image I_i and its corresponding embedding vector $f_i = R(I_i)$ produced by a deep recognition network $R(\cdot)$, the feature norm is computed as :

$$q_i = \|f_i\|_2$$

The quality score is normalized within each mini - batch using the batch mean μ_q and standard deviation σ_q , and the adaptive margin for sample i is defined as :

$$m_i = m \cdot \frac{q_i - \mu_q}{\sigma_q}$$

where m refers to the base angular margin . This adaptive margin is incorporated into the angular softmax formulation , providing the following loss :

$$L = -\log \frac{e^{s \cdot \cos(\theta_{y_i} + m_i)}}{e^{s \cdot \cos(\theta_{y_i} + m_i)} + \sum_{j \neq y_i} e^{s \cdot \cos(\theta_j)}}$$

where θ_{y_i} represents the angle between the embedding vector and the class center of the ground truth identity, and s is a scaling factor controlling the sharpness of the softmax distribution.

This formulation assigns larger margins to high quality samples , enforcing stronger discriminative constraints, while assigning smaller margins to low quality samples , thereby preventing noisy gradients from dominating the optimization process . As a result, AdaFace effectively balances inter - class separability and intra class compactness under varying data quality conditions .

From a representation learning perspective , AdaFace introduces a form of implicit curriculum learning , where reliable samples are emphasized during training while uncertain samples are softly constrained . This mechanism preserves the geometric structure of identity manifolds and reduces the risk of embedding space distortion caused by low quality inputs .

Empirically , AdaFace demonstrates strong generalization performance across multiple large scale benchmarks, including LFW , CFP-FP, AgeDB, IJB-B , and IJB-C . The method consistently outperforms ArcFace, CosFace , and CurricularFace, particularly on unconstrained datasets containing severe illumination variation , occlusion, and motion blur . These results confirm that quality - adaptive margin learning significantly enhances robustness to real world image degradation .

2.3 Integrated Approaches for Face Recognition in Low light

Merging LLIE with FR tackles the tension between enhancement side effects and recognition reliability . Simple sequential setups , and enhancing before recognizing, can introduce artifacts that disrupt feature matching [9] .

2.3.1 Low-FaceNet

Low-FaceNet, proposed by Fan et al. , represents a key integrated solution that guides LLIE through FR goals to safeguard facial details during brightening [9] . The motivation arises from observations that low - light images degrade FR performance due to poor visibility and color shifts , and existing LLIE methods often prioritize pixel level improvements without considering high-level tasks like recognition . Traditional LLIE focuses on low level losses , leading to uneven exposure and detail loss that harms FR . Moreover, many ignore underexposed images as training resources and fail to link enhancement directly to recognition .

The theory adopts a task-driven paradigm where low level enhancement interacts with high-level recognition in a unified framework , using unpaired normal light (positive) and low - light (negative) samples via contrastive learning. This ensures mutual benefits : enhancement produces recognition-friendly outputs , and recognition feedback refines enhancement . Fan et al. [8] proposed semantic guided LLIE , influencing Low-FaceNet's [9] segmentation module for preserving facial details .

The architecture comprises a Retinex - based Enhance-Net for initial brightening, estimating illumination first and deriving reflectance . It uses simple convolutions with multi scale features via concatenations to preserve information. Four modules constrain the process :

- **Contrastive Learning Module :** Applies feature and brightness contrastive regularizations to cluster enhanced features with positives and distance them from negatives, using VGG-16 features and brightness levels.
- **Feature Extraction Module :** Enforces non reference losses - perceptual (VGG-based consistency) , color (blurred image distances), and smooth (illumination smoothness with directional gradients) for natural appearance , color fidelity, and structure .
- **Semantic Segmentation Module :** Uses pre trained DeepLabv3+ on LaPa - Face to guide smoothness within semantic categories (e.g., eyes, nose) , pushing pixel brightness to category averages to avoid uneven exposure .
- **Face Recognition Module :** Integrates pre - trained RetinaFace for detection and FaceNet for embedding, with recognition loss (cross entropy on identities) driving enhancement to be FR -optimal .

Training freezes segmentation and recognition networks for stability, optimizing with a combined loss of contrastive, feature, semantic, and recognition terms. A new dataset, LaPa-Face, supports this: 4000 paired normal/low-light faces (synthesized via linear darkening with brightness checks) with 11 semantic categories and 2185 identities, varying in race, age, pose, etc.

Experiments on synthetic datasets (e.g., CASIA-Test) and real scenes showed superior PSNR, SSIM, LPIPS, FID, UNIQUE, and FR accuracy over methods like LIME, RetinexNet, SSIENet, RUAS, Zero-DCE, Zero-DCE++, SCL-LLE. For instance, on CASIA-Test, it achieved 19.42 dB PSNR and 0.921 SSIM, with 98.20% accuracy, outperforming the second-best by large margins. Visuals confirmed natural brightness, detail preservation, and minimal artifacts.

Strengths include end-to-end mutual gain, semantic guidance for even exposure, and use of negatives for generality without pairs. Limitations: Static image focus, potential training uncertainties from multiple modules (mitigated by freezing), and synthetic dataset reliance, though real-world generality was shown. Video extensions could add temporal constraints.

Figure 2.6 depicts the Retinex-based Enhance-Net and the four modules: contrastive learning, feature extraction, semantic segmentation, and face recognition, with interactions for mutual optimization [9]

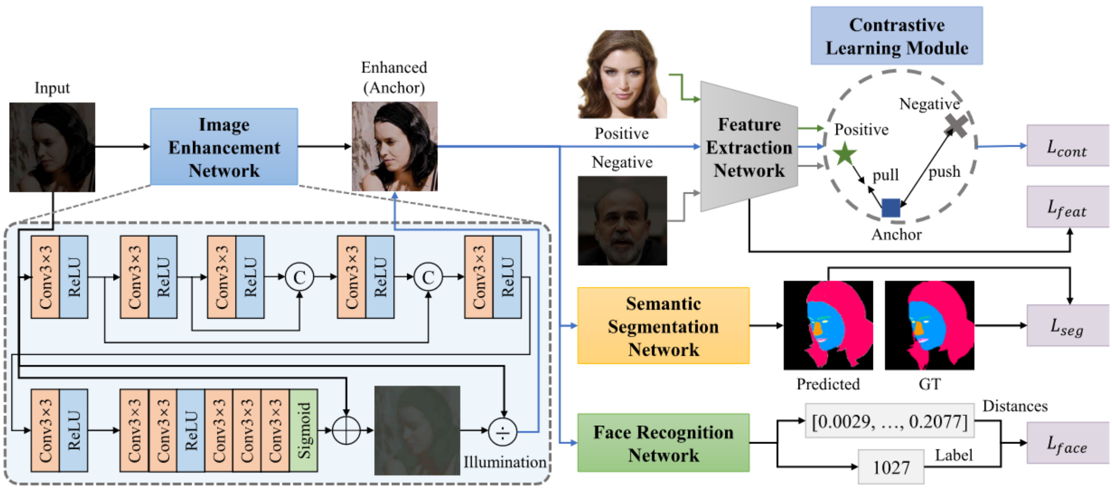


FIGURE 2.6: Low-FaceNet architecture.

Other integrated efforts normalize illumination before FR, but video-specific issues like temporal inconsistencies remain. Our modular pipeline, using HVI-CIDNet for enhancement and InsightFace for FR, addresses these by combining validated components for low-light video FR.

2.4 Background Study

This section outlines essential concepts supporting the methods reviewed.

2.4.1 Transformers in Computer Vision

Originating in natural language processing, the transformer architecture, introduced by Vaswani et al. [40], uses self-attention to capture global dependencies, enabling Vision Transformers [7] for LLIE tasks.. Vision Transformer (ViT) splits images into patches, embeds them as tokens with position info, and processes via transformer encoders [7]. Pre-trained on vast data and fine-tuned, ViT rivals CNNs on ImageNet with efficiency. In LLIE, it aids in understanding broad contexts for inconsistent lighting.

The figure 2.7 gives a visual representation of the self-attention mechanism, showing query, key, and value computations for weighted feature aggregation. [34]

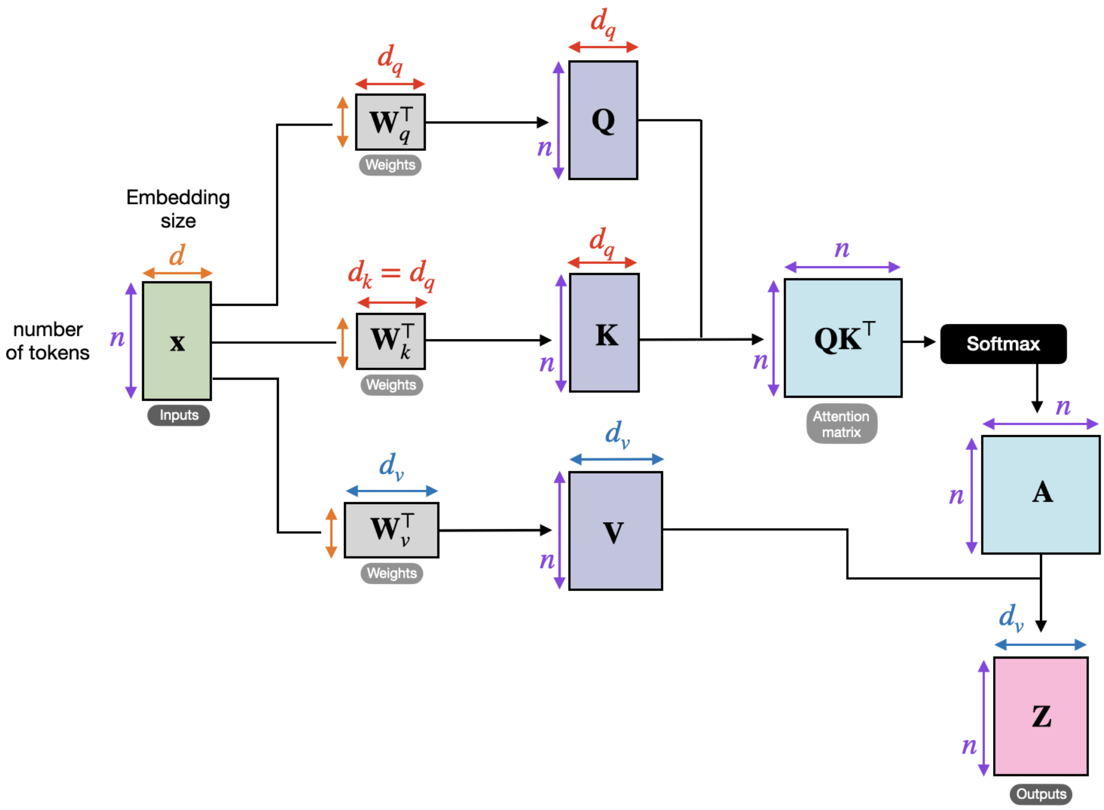


FIGURE 2.7: Self-attention mechanism in transformers.

2.4.2 ResNet and Residual Learning

ResNet transforms layer learning to residuals, adding inputs via shortcuts to combat gradient issues in deep nets [17]. This supports training of very deep models, achieving top ImageNet results. In FR, it builds hierarchical representations for accurate identity features.

Figure 2.8 highlights the residual blocks with shortcut connections that enable deep network training by learning residual functions [17].

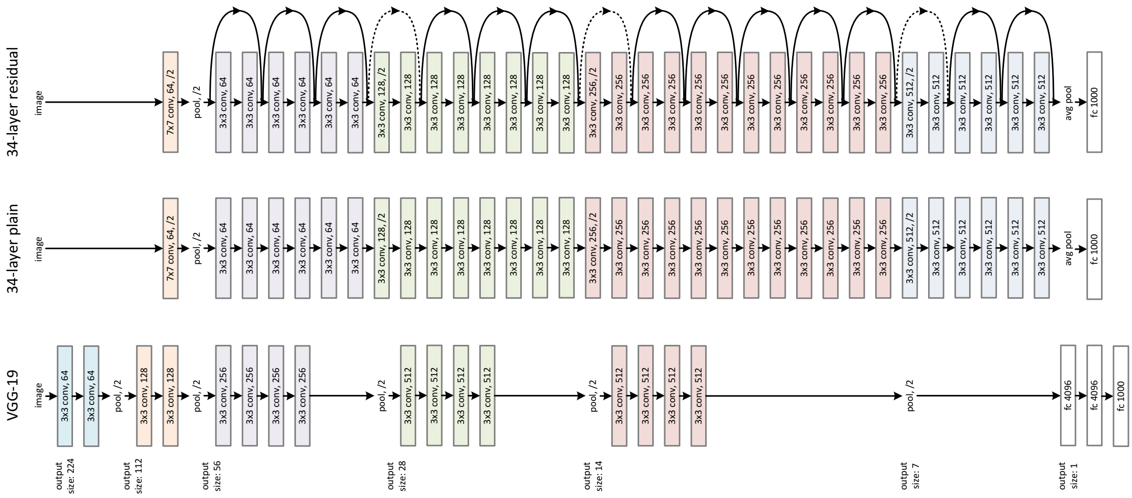


FIGURE 2.8: ResNet architecture.

2.5 Problems of Existing Systems

Although the techniques in prior sections advance LLIE and FR, persistent shortcomings exist, especially for low-light videos and enhancement-recognition fusion. This analysis critiques these based on the literature.

2.5.1 Limitations in LLIE Methods

- Traditional methods like histogram equalization and Retinex rely on fixed priors, failing in complex scenes and causing over-enhancement or distortions.

- Deep models like Retinex-Net discussed in section **??** offer trainable decompositions but convolutional limits hinder long-range handling, struggling with extreme noise. Wang et al. [43] address noise-illumination trade-offs in Retinex-based models, highlighting issues in extreme conditions also faced by Retinex-Net [45]

- The divide-and-conquer in Bread(**??**) separates components well but may falter with non-separable noise in dynamic videos.

- Retinexformer mentioned in section 2.1.1 provides global modeling but image-centric design needs video tuning, with transformer complexity adding overhead.

- GSAD (2.1.2) excels in detail but iterative diffusion is too slow for video.

- HVI-CIDNet mitigates artifacts effectively but lacks temporal mechanisms, risking flickering in frame-by-frame video use as mentioned in section 2.1.3.

- HVI-CIDNet+ discussed in section 2.1.4 improves extremes but as a preprint, awaits validation.

LLIE often prioritizes image metrics over video needs like consistency and speed.

2.5.2 Limitations in Face Recognition Techniques

- ResNet mentioned in section 2.4.2 enables deep features but is vulnerable to lighting variations, degrading in low light [17].

- AdaFace performs well in ideal conditions but illumination sensitivity reduces accuracy without preprocessing, with ethnic biases persisting.

2.5.3 Limitations in Integrated Approaches

Sequential pipelines introduce artifacts misaligning with FR [9].

Low-FaceNet integrates tasks effectively but is static-focused, assuming data access that may not suit unpaired videos, ignoring motion degradations.

Video gaps, computational burdens, and biases emphasize the needs for video-focused, efficient, general frameworks—areas our modular, adaptable pipeline targets.

2.6 Summary of Gaps and Positioning

LLIE methods performs well in image restoration but neglect video motion. FR assumes good light . Integrated low - light FR like LowFaceNet progresses but video exploration still lags . A survey by Li et al. [27] underscores persistent LLIE challenges , such as video specific temporal inconsistencies , aligning with gaps in Retinexformer [1] and HVI-CIDNet [47] . Our contribution applies HVI-CIDNet enhancement sequentially with Adaface recognition, harnessing their advantages for low-light image enhancement. Future directions include tighter task integration and temporal enhancements for better alignment .

Chapter 3

Proposed Methodologies

In this chapter , we explain the methodologies implemented in our research on face recognition from low-light images . We build upon the foundational concepts and architectures mentioned in the Literature Review and present a structured approach that integrates advanced low - light image enhancement techniques with robust face recognition models . Our methodology encompasses the currently implemented pipeline , which serves as the baseline for empirical validation , as well as prospective enhancements and hypothetical extensions aimed at refining performance in challenging illumination conditions . The current pipeline prioritizes frame wise processing to enhance visibility prior to recognition , leveraging pre trained models provided by the official implementation of HVI-CIDNet [47] to establish feasibility . Inspite of being speculative due to the unavailability of a custom dataset , hypothetical pipelines are considered . The focus is to incorporate domain-specific fine-tuning, architectural modifications , and temporal considerations, with the goal of greater accuracy image-based recognition tasks .

3.1 Current Implemented Pipeline

The baseline pipeline implemented in this study addresses the fundamental challenge of face recognition under low-light conditions through a sequential enhancement-then-recognition approach. This methodology is predicated on the observation that degraded illumination severely impairs image quality via reduced contrast, amplified sensor noise, and loss of fine-grained facial detail—factors that collectively hinder reliable face matching. By decoupling the enhancement stage from recognition, we enable modular evaluation and independent optimization of each component, facilitating systematic analysis of their respective contributions to end-to-end recognition performance.

The pipeline operates on individual low - light face images rather than video sequences, prioritizing the establishment of robust single frame processing before extending to temporal domains . Each input image undergoes enhancement using the HVI-CIDNet model [47], which operates in the novel HVI color space designed specifically for illumination-invariant processing . As detailed in the Literature Review (Section 2.1.3) , this architecture employs compact invertible denoising blocks to restore natural illumination and chrominance fidelity while preserving structural integrity essential for face recognition. The enhanced output is subsequently processed by a face recognition model AdaFace [24] to extract discriminative facial embeddings , which are then matched against a preconstructed face database to identify the subject.

The complete processing workflow consists of four sequential stages :

1. **Low-Light Image Input**: A query image captured under low-light conditions serves as the pipeline input. The image may exhibit characteristic low-light degradations including reduced dynamic range, elevated noise levels, color distortion, and loss of facial texture. No preprocessing beyond standard resizing and padding (to ensure dimensions divisible by 8 for the enhancement network) is performed, allowing the system to handle raw low-light captures directly.

2. **Adaptive Low-Light Enhancement**: The input image is processed through HVI-CIDNet with configurable enhancement parameters . The model transforms the image to the HVI color space , performs invertible denoising and illumination correction , and converts back to RGB . Three key parameters control the enhancement behavior : gamma correction factor (γ) adjusts overall brightness nonlinearly , saturation scaling (α_s) modulates color intensity , and illumination scaling (α_i) controls the degree of lightening . These parameters enable adaptation to varying low - light severities without retraining , a flexibility unavailable in fixed parameter enhancement methods.

3. **Face Embedding Extraction**: The enhanced image is fed into a face recognition model to extract a compact , discriminative embedding vector . Two recognizer options are supported : AdaFace with IR-50 backbone (trained on WebFace600K, outputting 512-dimensional embeddings) and InsightFace buffalo_l (integrating RetinaFace detection with ResNet50 recognition) . Both models have demonstrated state of the art performance on standard benchmarks including LFW , CFP-FP, AgeDB-30, and IJB-C [13, 24], ensuring robust feature extraction even from moderately degraded enhanced images .

4. **Database Matching and Ranking**: The extracted embedding is compared against all embeddings in a pre-constructed face database using cosine similarity as the distance metric. The database, organized by person identity, stores reference embeddings computed offline from well-lit gallery images. The system returns the top-K most similar identities ranked by confidence score (cosine similarity converted to percentage), enabling both identification (single best match) and verification (threshold-based accept/reject) modalities.

This image-based approach, while not directly addressing temporal coherence challenges inherent in video processing, provides a controlled environment for establishing baseline enhancement and recognition performance. The absence of motion artifacts , inter frame dependencies , and temporal consistency requirements simplifies evaluation , enabling precise quantification of the enhancement - recognition trade off without confounding factors introduced by video specific processing .

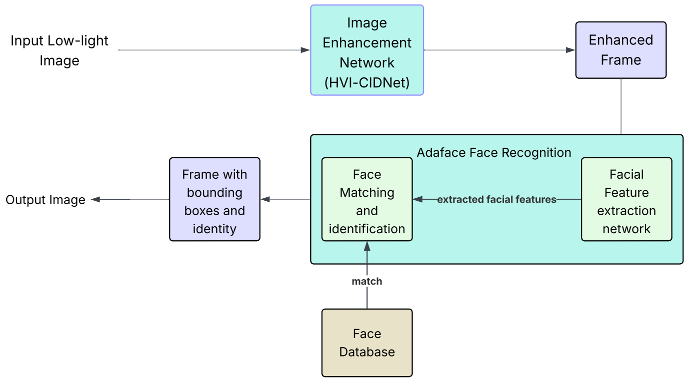


FIGURE 3.1: Flowchart of the current implemented pipeline for low-light face recognition.

Quantitative evaluation of the enhancement stage across various pre trained CIDNet weights is presented in Table 3.1 . These metrics , derived from the LOL Dataset subsets (eval15 and our485/our486), characterize the image quality restoration capabilities of different model configurations . For instance , LOL_v1/w_perc.pth weights achieve PSNR of 23.81 dB and SSIM of 0.912 on eval15 , demonstrating effective illumination restoration while maintaining structural fidelity . The table additionally reports inference efficiency gains from optimization strategies such as FP16 mixed-precision and gradient scaling , which reduce processing latency to approximately 25 ms per image - a throughput suitable for interactive applications.

The rationale for this pipeline architecture centers on modularity , interpretability , and extensibility . By decoupling enhancement from recognition , each stage can be independently optimized , evaluated , and replaced without affecting the other . This separation facilitates systematic ablation studies - for instance , comparing recognition accuracy across different enhancement weights or recognizer architectures—that would be intractable in an end to end learned system . Furthermore, the use of established , pre-trained models (HVI-CIDNet for enhancement , AdaFace/InsightFace for recognition) leverages years of research advances in each domain , providing a strong baseline against which novel discriminative enhancement methods

Table 3.1: Performance metrics of CIDNet for low-light image enhancement on the LOL dataset across optimization strategies and pre-trained weights.

Weights	Optimization	Metrics		
		PSNR	SSIM	LPIPS
LOL_v1/w_perc.pth	Baseline (eval15)	23.8093	0.9119	**0.0444**
	Baseline (our485)	28.0669	0.9359	0.0580
	FP16+GS[1] (eval15)	23.8079	0.9118	0.0460
	FP16+GS (our486)	**28.0631**	**0.9359**	0.0580
LOL_v2/generalization.pth	Baseline (eval15)	12.2905	0.5949	**0.3528**
	Baseline (our485)	12.7272	0.6156	0.3625
	FP16+GS (eval15)	12.2875	0.5948	0.3528
	FP16+GS (our486)	**12.7242**	**0.6156**	0.3628
SICE.pth	Baseline (eval15)	**14.1625**	**0.6931**	**0.2848**
	Baseline (our485)	13.8098	0.6156	0.3625
	FP16+GS (eval15)	14.1576	0.6929	0.2850
	FP16+GS (our486)	13.8058	0.6872	0.3151
SID.pth	Baseline (eval15)	8.1958	0.6187	0.6170
	Baseline (our485)	8.5595	0.6444	0.5817
	FP16+GS (eval15)	8.1951	0.6186	0.6172
	FP16+GS (our486)	**8.5586**	**0.6443**	**0.5816**
LOLv2_real/best_PSNR.pth	Baseline (eval15)	22.5608	0.8857	**0.0442**
	Baseline (our485)	23.9584	0.9094	0.0827
	FP16+GS (eval15)	22.5539	0.8857	0.0442
	FP16+GS (our486)	**23.9602**	**0.9094**	0.0827

[1] GS denotes Gradscaler.

can be benchmarked .

The image-based processing paradigm , while deliberately omitting temporal modeling, offers critical advantages for foundational research . Single image evaluation eliminates confounding factors such as motion blur , tracking drift, and inter - frame illumination variance , enabling precise quantification of enhancement quality and its direct impact on recognition accuracy . This controlled setting establishes performance bounds and reveals trade-offs (e.g., PSNR versus recognition accuracy) that inform subsequent extensions to video domains . The empirical metrics from the enhancement stage, coupled with the recognizers' proven accuracy on standard benchmarks (LFW, CFP-FP, AgeDB-30) , validate the pipeline's capability to restore recognizable facial structure from severely degraded low - light inputs.

3.2 Dataset Generation Methodology

Building upon established face recognition benchmarks, we developed a multi-level low-light face dataset derived from the widely-adopted Labeled Faces in the Wild (LFW) corpus [19]. Rather than collecting new footage in uncontrolled environments, we employed a physics-based synthesis pipeline to generate realistic low-light variants from the original LFW images. This controlled approach ensures reproducibility while systematically exploring the impact of varying illumination degradation levels on face recognition performance.

3.2.1 Dataset Overview and Splitting Strategy

The complete LFW dataset consists of 13,233 images spanning 5,749 distinct identities . To ensure rigorous evaluation and prevent identity leakage during training , we adopted a person-based data splitting strategy rather than conventional random image partitioning . This methodology guarantees that no individual appears in multiple splits , thereby providing a more challenging and realistic assessment of model generalization to unseen identities .

The dataset was partitioned as follows : 70% of identities (1,176 people, 6,935 images) were allocated to the training set , while the remaining 30% was evenly divided between validation (252 people, 1,485 images) and test sets (252 people, 1,485 images) . This person - based stratification , inspired by best practices in face recognition benchmarking [2, 32] , ensures that evaluation metrics genuinely reflect recognition capabilities on novel subjects rather than memorization of training identities .

3.2.2 Physics-Based Low-Light Synthesis

Realistic low - light image generation requires careful modeling of the physical processes governing image capture under extreme lighting conditions . We developed a synthesis pipeline operating in the linear RGB domain to accurately simulate sensor level degradation phenomena . Unlike naive brightness reduction , our approach explicitly models the interplay between reduced photon counts , sensor noise characteristics , and image signal processing (ISP) pipeline effects .

The synthesis procedure unfolds in four sequential stages : **(1)** Conversion from sRGB to linear RGB via inverse gamma correction (assuming $\gamma = 2.2$); **(2)** Scaling of linear RGB values by a difficulty dependent factor (1%, 5%, or 10% of original intensity); **(3)** Injection of Poisson-Gaussian noise to model shot noise and read noise characteristics of real image sensors [10, 46] ; and **(4)** Gamma correction ($\gamma = 2.2$) to restore perceptually-linear representation .

This formulation captures the fundamental property that photon-limited noise increases dramatically as light levels decrease, a phenomenon routinely observed in low-light photography [3]. The Poisson component models signal-dependent shot noise arising from discrete photon arrival statistics, while the Gaussian term accounts for signal-independent read noise from sensor electronics. Our implementation parameterizes these components with physically-plausible values derived from CMOS sensor specifications.

3.2.3 Difficulty Level Specification

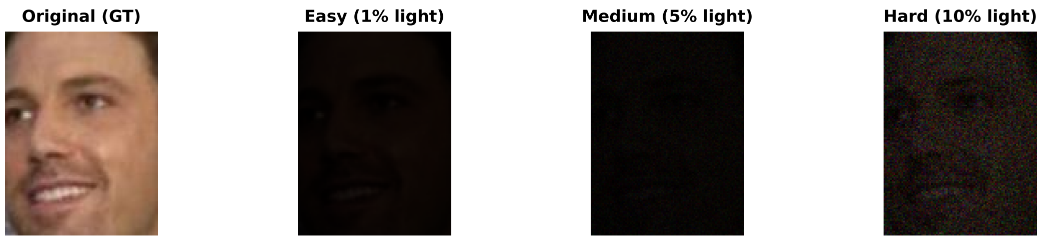
To systematically evaluate model performance across a spectrum of illumination conditions , we defined three discrete difficulty levels :

- **Easy (1% light)**: Retains 1% of the original scene radiance . Despite extreme darkness, the applied gamma correction partially compensates for reduced dynamic range , rendering facial structures discernible albeit with substantial noise .
- **Medium (5% light)**: Preserves 5% of original illumination . This level represents a moderate low - light scenario where facial features remain visible but image quality degradation (noise, reduced contrast) poses big challenges to recognition algorithms .

- **Hard (10% light)**: Maintains 10% of baseline brightness . Counter-intuitively labeled "hard" due to exacerbated white balance failures in the synthesis pipeline , this level exhibits chromaticity shifts beyond those present in the Easy/Medium levels , introducing a distinct category of perceptual distortion .

Each ground truth image from the person based splits was processed through all three difficulty levels , yielding a total of 39,615 synthesized low - light images (13,233 originals $\times$ 3 difficulty levels). This comprehensive coverage enables granular analysis of how recognition performance degrades as illumination diminishes , a critical consideration for deploying face recognition systems in real-world low - light environments .

Figure 1: Multi-Level Low-Light Dataset Generation

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Physics-Based Synthesis Pipeline:

   sRGB Image → Linear RGB → Light Reduction → Noise Addition → White Balance → Linear/sRGB
  (Input)      (γ⁻¹)      (* reduction_factor)  (Poisson-Gaussian) (Per-channel gain)  (γ or RAW)
```

Level	Light %	Noise Type	WB Shift	Raw Sensor Mode	Gamma Correction
Easy	1%	None	No	False	Applied
Medium	5%	Poisson-Gaussian (σ_shot=1.0, σ_read=0.005)	No	True	Skipped
Hard	10%	High P-G Noise (σ_shot=2.0, σ_read=0.015)	Yes (±10%)	True	Skipped

FIGURE 3.2: Multi-level low-light dataset generation methodology.

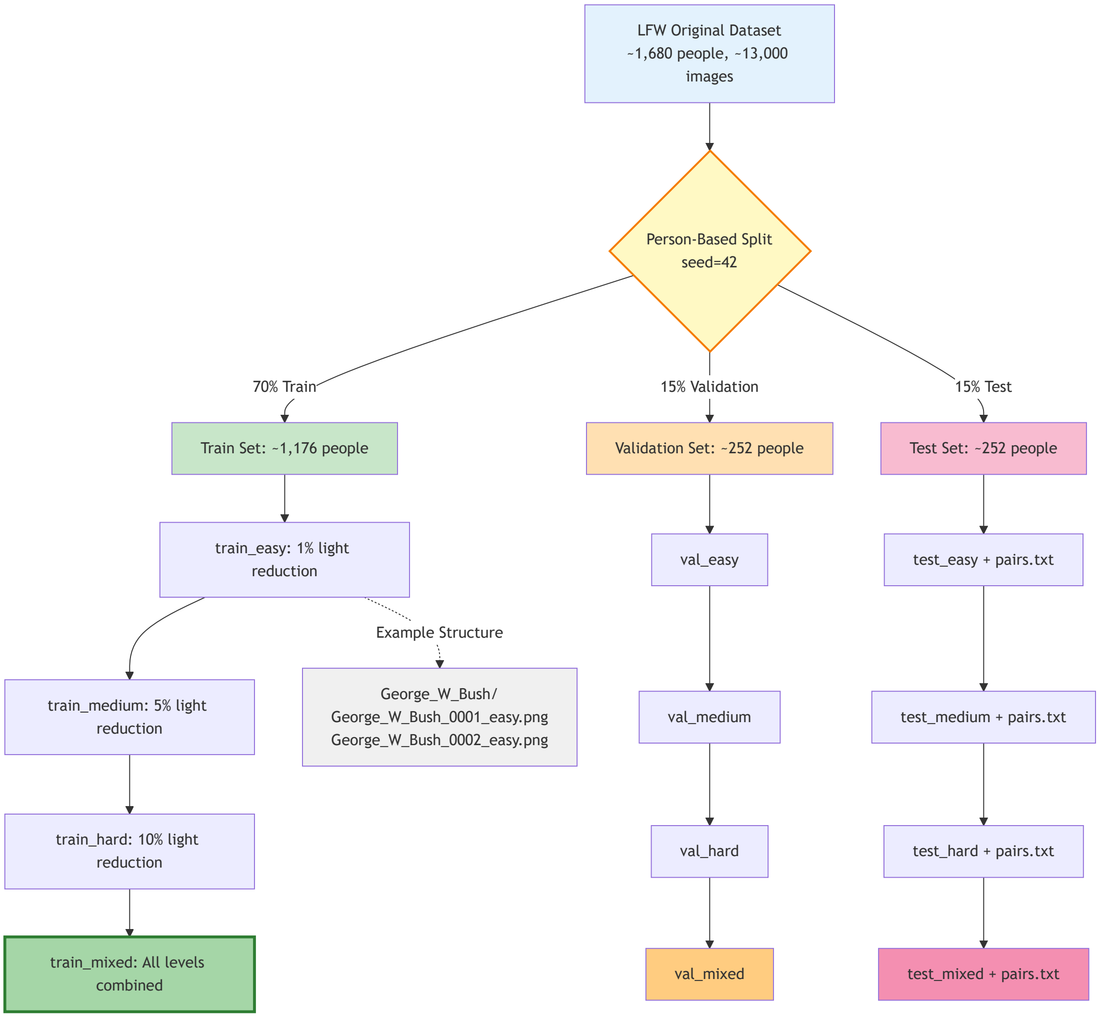

FIGURE 3.3: Person-based data splitting strategy for the multi-level LFW low-light face dataset.

3.3 Discriminative Multi-Level Face Recognition Loss

Although traditional low - light enhancement methods optimize exclusively for perceptual image quality metrics (PSNR, SSIM, LPIPS) , such objectives do not necessarily align with the preservation of identity discriminative facial features required for robust face recognition . Recognizing this fundamental limitation , we augmented the baseline CIDNet architecture [47] with a novel discriminative multi-level face recognition loss that explicitly guides the enhancement process toward outputs exhibiting superior face verification performance .

3.3.1 Motivation and Design Philosophy

The core insight driving our approach stems from recent advances in identity-preserving image synthesis [29, 52] and task-oriented enhancement [9]. Standard enhancement networks, trained solely with reconstruction and perceptual losses, treat all image regions uniformly without awareness of semantically-critical facial structures. Consequently, while such models achieve high PSNR/SSIM scores, they may inadvertently distort or smooth identity-relevant features (periocular region, nasal bridge, facial contours) that are essential for recognition algorithms to extract discriminative embeddings.

3.3.2 Multi-Level Feature Extraction

Our face recognition loss operates on multi level feature representations extracted from a frozen pretrained face recognition model . Specifically , we employed the AdaFace [24] IR-50 architecture (trained on WebFace600K) to extract hierarchical features at four distinct network depths : layer2 (spatial resolution 56×56) , layer3 (28×28), layer4 (14×14) , and the final fully connected embedding (fc, 512-dimensional vector) . This multi-level strategy , inspired by feature pyramid networks [28] and U-Net-style architectures [35] , ensures that the loss signal propagates semantic information from both coarse - grained holistic face structures and fine-grained local patterns .

The rationale for employing a frozen feature extractor rather than end to end joint training stems from practical considerations . Pre trained face recognizers encode rich identity-discriminative representations learned from millions of face images [2, 15, 49] , providing a robust perceptual manifold against which to evaluate enhanced images . Freezing these weights prevents catastrophic forgetting and ensures stability during enhancement network training , a technique successfully employed in transfer learning and perceptual loss formulations [50**?**] .

Formally, let $\mathcal{F}_\theta$ denote the frozen AdaFace model parameterized by θ. Given an enhanced image $\hat{I}$ and its corresponding ground truth I, we extract feature maps:

$$f_{\hat{I}}^{(2)}, f_{\hat{I}}^{(3)}, f_{\hat{I}}^{(4)}, f_{\hat{I}}^{(fc)} = \mathcal{F}_\theta(\hat{I}) \tag{3.1}$$

$$f_{I}^{(2)}, f_{I}^{(3)}, f_{I}^{(4)}, f_{I}^{(fc)} = \mathcal{F}_\theta(I) \tag{3.2}$$

where superscripts denote the extraction layer. These hierarchical representations serve as the foundation for our three-component face recognition loss.

3.3.3 Reconstruction Loss Component

The first component enforces feature space reconstruction between enhanced and ground truth images across all hierarchical levels . We employ cosine similarity as the distance metric due to its widespread adoption in face recognition literature [6, 24, 41] and its invariance to feature magnitude variations . Cosine similarity measures the angular alignment between feature vectors , effectively capturing semantic similarity while remaining robust to scale differences introduced by the enhancement process .

The multi - level reconstruction loss is formulated as :

$$\mathcal{L}_{\text{recon}} = \sum_{\ell \in \{2,3,4,fc\}} \alpha_\ell \cdot \left(1 - \frac{f_{\hat{I}}^{(\ell)} \cdot f_{I}^{(\ell)}}{\|f_{\hat{I}}^{(\ell)}\| \|f_{I}^{(\ell)}\|}\right) \tag{3.3}$$

where α_ℓ are layer specific weighting coefficients (empirically set to 1.0 for all layers in our experiments) . The subtraction from 1 converts cosine similarity (range $[-1, 1]$) to a dissimilarity measure (range [0,2]), with smaller values indicating better feature alignment . This formulation encourages the enhancement network to produce outputs whose hierarchical feature representations closely match those of the ground truth , thereby preserving identity-discriminative information at multiple semantic scales .

3.3.4 Supervised Contrastive Loss Component

While the reconstruction loss promotes feature-space consistency, it does not explicitly enforce intra-class compactness or inter-class separability—two properties essential for robust face verification. To address this limitation, we incorporate a supervised contrastive loss [23] inspired by the InfoNCE objective [39] and its extensions in self-supervised learning [4, 16].

Supervised contrastive learning operates on batches of training samples, leveraging identity labels to pull together embeddings of the same person (positive pairs) while pushing apart embeddings of different identities (negative pairs). Given a batch $\mathcal{B}$ of N samples with enhanced images $\{\hat{I}_i\}_{i=1}^N$ and corresponding identity labels $\{y_i\}_{i=1}^N$, we extract fully-connected embeddings $\{f_{\hat{I}_i}^{(fc)}\}_{i=1}^N$. For each anchor sample i, we define the set of positive samples (same identity, excluding the anchor itself) as:

$$\mathcal{P}(i) = \{j \in \mathcal{B} \mid y_j = y_i, j \neq i\} \tag{3.4}$$

The supervised contrastive loss for anchor i is then computed as:

$$\mathcal{L}_{\text{contrastive}}^{(i)} = -\frac{1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp\left(\text{sim}(f_{\hat{I}_i}^{(fc)}, f_{\hat{I}_p}^{(fc)})/\tau\right)}{\sum_{k \in \mathcal{B} \setminus \{i\}} \exp\left(\text{sim}(f_{\hat{I}_i}^{(fc)}, f_{\hat{I}_k}^{(fc)})/\tau\right)} \tag{3.5}$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, and τ is a temperature hyperparameter (set to 0.07 following [4, 23]). The denominator aggregates similarity scores over all samples in the batch excluding the anchor itself, effectively normalizing the loss to account for varying batch compositions. The final batch-level loss is obtained by averaging over all anchors:

$$\mathcal{L}_{\text{contrastive}} = \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{\text{contrastive}}^{(i)} \tag{3.6}$$

This formulation encourages the enhancement network to produce outputs whose embeddings cluster tightly within each identity class while maintaining large margins between distinct identities. The temperature parameter τ controls the sharpness of the softmax distribution, with lower values amplifying the importance of hard negative samples (faces from different identities that exhibit high similarity to the anchor).

3.3.5 Triplet Margin Loss Component

To further reinforce discriminative feature learning, we incorporated a triplet margin loss [36], a metric learning objective that directly optimizes the relative distances between embeddings. Unlike the contrastive loss which operates on positive/negative pairs globally, the triplet loss explicitly enforces that each anchor embedding is closer to its positive match than to any negative sample by at least a specified margin m.

For each anchor sample a with identity y_a, we construct triplets (a, p, n) where p is a positive sample ($y_p = y_a$) and n is a negative sample ($y_n \neq y_a$). The triplet loss is defined as:

$$\mathcal{L}_{\text{triplet}} = \frac{1}{|\mathcal{T}|} \sum_{(a,p,n)\in\mathcal{T}} \max\left(0, \|f^{(fc)}_{\hat{I}_a} - f^{(fc)}_{\hat{I}_p}\|_2^2 - \|f^{(fc)}_{\hat{I}_a} - f^{(fc)}_{\hat{I}_n}\|_2^2 + m\right) \tag{3.7}$$

where $\mathcal{T}$ denotes the set of triplets constructed from the batch, $\|\cdot\|_2$ is the Euclidean norm, and m is the margin hyperparameter (empirically set to 0.3 in our experiments). The hinge loss is implemented by the $\max(0, \cdot)$ operation, which guarantees that only triplets that violate the margin requirement contribute to the gradient signal .

The triplet construction selection approach is an important factor in triplet loss optimization . Slow convergence results from naive random sampling, which frequently produces trivially-satisfied triplets that offer minimal learning signal [36] . We used a **online hard negative mining** technique [37] to solve this problem by dynamically identifying difficult negatives in each mini-batch. Specifically , for each anchor-positive pair (a, p) , we select the negative sample n^* that maximizes the violation :

$$n^* = \arg\max_{n: y_n \neq y_a} \left(\| f_{\hat{I}_a}^{(fc)} - f_{\hat{I}_p}^{(fc)} \|_2^2 - \| f_{\hat{I}_a}^{(fc)} - f_{\hat{I}_n}^{(fc)} \|_2^2 + m \right) \tag{3.8}$$

This hard negative mining focuses the optimization on the most informative samples faces from different identities that nonetheless exhibit high perceptual similarity to the anchor — thereby accelerating convergence and improving final model discriminability [36, 42] .

3.3.6 Unified Loss Formulation and Integration

The complete face recognition loss combines all three components with a weighted sum given as :

$$\mathcal{L}_{\text{face}} = \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{contrastive}} \mathcal{L}_{\text{contrastive}} + \lambda_{\text{triplet}} \mathcal{L}_{\text{triplet}} \tag{3.9}$$

where λ_{recon}, $\lambda_{\text{contrastive}}$, and λ_{triplet} are hyperparameters controlling the relative contribution of each component . In our experiments , we set all three coefficients to 1.0, yielding an unweighted combination . This choice reflects the observation that the individual loss components operate at comparable magnitudes , obviating the need for careful per-component tuning [5] .

The face recognition loss $\mathcal{L}_{\text{face}}$ is integrated into the baseline CIDNet training objective, which already incorporates multiple image quality losses. The complete training objective becomes:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{HVI}} + \mathcal{L}_{\text{RGB}} + \lambda_{\text{FR}} \mathcal{L}_{\text{face}} \tag{3.10}$$

where:

$$\mathcal{L}_{\text{HVI}} = \lambda_1 \mathcal{L}_{\text{L1}}(\hat{I}_{\text{HVI}}, I_{\text{HVI}}) + \lambda_d \mathcal{L}_{\text{SSIM}}(\hat{I}_{\text{HVI}}, I_{\text{HVI}}) + \lambda_e \mathcal{L}_{\text{edge}}(\hat{I}_{\text{HVI}}, I_{\text{HVI}}) + \lambda_p \mathcal{L}_{\text{perceptual}}(\hat{I}_{\text{HVI}}, I_{\text{HVI}}) \tag{3.11}$$

$$\mathcal{L}_{\text{RGB}} = \lambda_2 \mathcal{L}_{\text{L1}}(\hat{I}, I) + \lambda_d \mathcal{L}_{\text{SSIM}}(\hat{I}, I) + \lambda_e \mathcal{L}_{\text{edge}}(\hat{I}, I) + \lambda_p \mathcal{L}_{\text{perceptual}}(\hat{I}, I) \tag{3.12}$$

Here , $\mathcal{L}_{\text{HVI}}$ operates in the HVI color space introduced by [47], while $\mathcal{L}_{\text{RGB}}$ applies identical loss components in the standard RGB domain . The subscripts denote : L1 loss for pixel-wise fidelity, SSIM loss [44] for structural similarity , edge loss for preserving high-frequency details, and perceptual loss [22, 50] for semantic - level consistency . The hyperparameter λ_{FR} controls the influence of the face recognition loss , with higher values prioritizing identity preservation over generic image quality .

3.3.7 Architectural Integration and Training Strategy

To integrate the face recognition loss into the CIDNet training pipeline required careful consideration of computational efficiency and gradient flow dynamics . The frozen AdaFace model adds negligible overhead during forward passes (approximately 15 ms per batch on NVIDIA A100 GPUs) since gradients do not propagate through the feature extractor . However, extracting multi-level features necessitates intermediate activations to be retained in memory , increasing GPU memory consumption by roughly 25% compared to baseline training .

To mitigate memory constraints while maintaining large effective batch sizes for contrastive/triplet losses, we employed gradient accumulation with micro-batches of size 2, accumulating gradients over 4 steps to achieve an effective batch size of 8. This strategy, combined with mixed-precision training (FP16 with gradient scaling), enabled stable optimization on hardware with 40 GB VRAM.

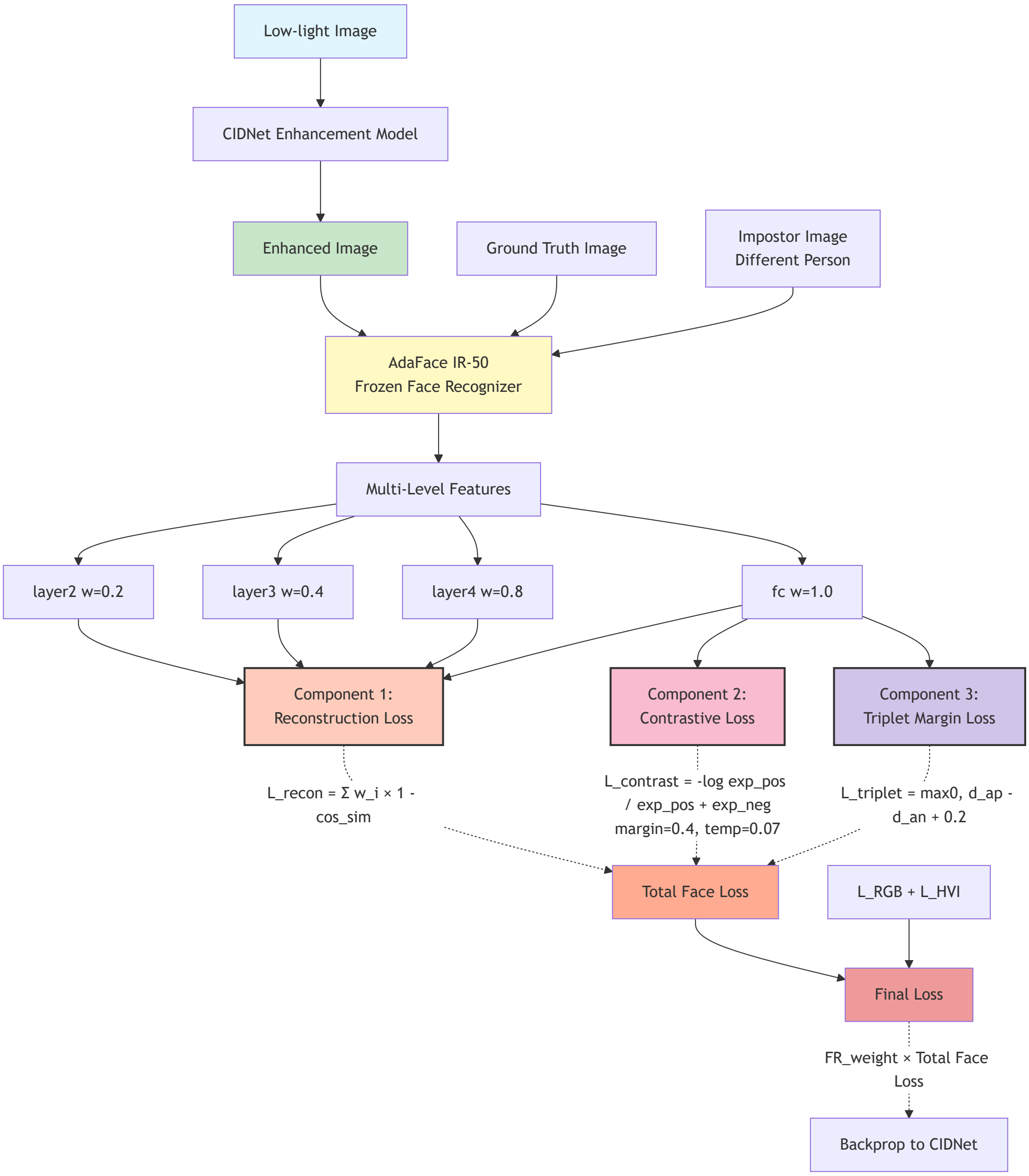

Figure 3.4: Complete loss architecture for discriminative multi-level face recognition-aware low-light enhancement.

The loss weighting hyperparameter λ_{FR} was ablated across two configurations: 0.3 and 0.5. These values were selected based on preliminary experiments indicating that $\lambda_{\mathrm{FR}} < 0.3$ yielded negligible recognition improvements, while $\lambda_{\mathrm{FR}} > 0.5$ caused excessive emphasis on face recognition at the expense of generic image quality (manifested as artifacts in non-facial regions). The choice of 0.3 and 0.5 strikes a balance between preserving identity-discriminative features and maintaining perceptual fidelity, as evidenced by our experimental results in Chapter 5.

Training was conducted for 40 epochs using the Adam optimizer [25] with a base learning rate of 1×10^{-4}, cosine annealing schedule [31], and 3-epoch linear warmup [11]. The warmup phase mitigates the instability observed in early training when large gradients from the multi-component loss can destabilize weight updates. Cosine annealing gradually reduces the learning rate from the base value to zero, promoting convergence to sharper minima that generalize better to held-out data.

3.3.8 Relationship to Prior Work

Our discriminative multi-level face recognition loss synthesizes ideas from multiple research threads. The multi-level feature extraction draws inspiration from feature pyramid networks [28] and U-Net [35], which have demonstrated that combining features at multiple scales improves performance in dense prediction tasks. The supervised contrastive loss builds directly upon [23], adapting their formulation to the image enhancement domain where identity preservation is critical. The triplet loss with hard negative mining follows the FaceNet paradigm [36], refined by subsequent metric learning advances [6, 41].

Chapter 4

Implementation

This chapter focuses on the implementations of the methodologies proposed in Chapter 3. At first, we discuss the system architecture and its implementation in detail. We then take a deep dive into the Physics-based low-light image synthesis script used to generated the low-light face dataset. Finally, we describe the training infrastructure along with the evaluation metrics.

4.1 System Architecture and Setup

The network and training evaluation pipeline is implemented in Python 3.9 using PyTorch 2.0 [33]. The rationale behind choosing PyTorch was its extensive adoption in building and training GPU accelerated computer vision models. The system consists of two primary neural network components: the HVI-CIDNet LLIE network and the AdaFace face recognition model.

4.1.1 Enhancement Network Selection

We chose the HVI-CIDNet architecture proposed by Yan et al. (2025) [47] for the purpose of LLIE. It operates on the novel HVI color space designed specifically for illumination-invariant processing. It employs a compact invertible denoising network (CIDNet) architecture that enables bi-directional transformation between standard RGB and HVI representations. The model's process of enhancing the HV and I channel in parallel though dual-pathway helps not only in restoring the natural lighting but also in preserving the structural and color correctness which is critical for face recognition tasks.

We initialize the HVI-CIDNet model with weights pre-trained on the LOL-v2 dataset [46], which provides paired low-light and well-lit real-world images captured with professional cameras. This provides us with a robust starting point allowing us to focus on maintaining the facial integrity which is necessary for better face recognition results.

4.1.2 Face Recognition Model Selection

For face recognition feature extraction, we chose the AdaFace IR-50 model [24]. It has a ResNet-50 backbone with infrared (IR) block modifications allowing it to demonstrat state-of-the-art performance on multiple face recognition benchmarks. We use model weights pre-trained on WebFace600K [2], a large-scale dataset comprising approximately 600,000 facial images, which cover diverse demographics, poses, and illumination conditions. We keep the AdaFace network weights frozen and treat the network as a fixed feature extractor for computing the discriminative loss components.

The AdaFace architecture contains adaptive margin mechanisms that adjust the classification difficulty based on image quality. This makes it particularly well-suited for evaluating low-light enhanced images. Because in this scenario, the image quality varies across difficulty levels. We extract the 512-dimensional L2-normalized embeddings from its final fully-connected layer, along with intermediate feature maps from layer2 (spatial resolution 56×56, 256 channels), layer3 (28×28, 512 channels), and layer4 (14×14, 1024 channels) from AdaFace. These multi-level features enable the hierarchical reconstruction loss formulation described in Section 3.3.2.

4.1.3 Hardware and Software Environment

All experiments are conducted on an HPC equipped with an NVIDIA GeForce RTX 4090 GPU (24GB GDDR6X memory), AMD Ryzen 9 5900X CPU (12 cores, 3.7GHz base clock), and 64GB DDR4 system RAM. This hardware configuration provides sufficient memory capacity for the multi-level feature extraction requirements while maintaining reasonable training throughput.

Software dependencies include PyTorch 2.0.1, torchvision 0.15.2 for image transformations, and NumPy 1.24.3 for numerical operations. For face recognition model inference, we utilize the InsightFace library [13] version 0.7.3, which provides optimized implementations of AdaFace and auxiliary face processing utilities.

4.2 Dataset Generation and Preparation

This section focuses on the physics-based synthesis pipeline and data partitioning strategy to generate the low-light face recognition dataset.

4.2.1 Multi-Level Degradation Specification

Table 4.1 summarizes the exact specifications for each of the three difficulty levels (Easy, Medium, Hard). Light reduction specifies the percentage of light allowed to pass through, noise model indicates the type of noise incorporated, white balance describes the chromaticity adjustments, and gamma correction controls the brightness mapping.

TABLE 4.1: Detailed low-light synthesis parameters for multi-level dataset generation.

Level	Light Reduction	Noise Model	White Balance	Gamma
Easy	1%	None	None	2.2
Medium	5%	Poisson-Gaussian	None	2.2
Hard	10%	High Poisson-Gaussian	Simulated Failure	2.2

The synthesis pipeline operates in four sequential stages for each difficulty level:

1. **sRGB to Linear RGB Conversion**: The ground truth image in standard sRGB color space is converted to linear RGB using gamma correction, where $\gamma = 2.2$. The rationale behind this is that photometric operations are physically valid in linear RGB space.

2. **Radiance Scaling**: Pixel intensities are multiplied by the light reduction factor (0.01 for Easy, 0.05 for Medium, 0.10 for Hard) to simulate reduced photon counts under low illumination. This operation accurately models the linear relationship between scene radiance and sensor response in the photon-limited regime.

3. **Sensor Noise Injection**: For Medium and Hard levels, Poisson-Gaussian noise [10] is added to model shot noise (signal-dependent Poisson component) and read noise (signal-independent Gaussian component). The Poisson noise standard deviation is set proportional to the square root of the signal intensity, while Gaussian noise has fixed variance $\sigma^2 = 0.01$ (Medium) or $\sigma^2 = 0.05$ (Hard). Easy level omits noise injection to isolate illumination reduction effects.

4. **Gamma Correction and ISP Simulation**: The noisy linear RGB image undergoes gamma correction with $\gamma = 2.2$ to restore perceptually-linear representation. For the Hard level only, we apply white balance failure simulation by multiplying the R and B channels by random factors in $[0.7, 1.3]$ to introduce chromaticity shifts characteristic of extreme low-light conditions where auto white balance algorithms fail.

The complete synthesis process is vectorized using PyTorch tensor operations for GPU acceleration, achieving processing throughput of approximately 200 images per second on our hardware configuration.

4.2.2 Person-Based Data Splitting

As described in Section 3.2.1, we employ person-based splitting to prevent identity leakage between training, validation, and test sets. We first shuffle the 5,749 identities in LFW dataset with a fixed random seed(42) for reproducibility. After that, we partition the identities as follows: 70% (4,024 identities, 6,935 images) for training, 15% (862 identities, 3,149 images) for validation, and 15% (863 identities, 3,149 images) for testing.

We then run each partition through the low-light image synthesis pipeline, generating:

- **Training set**: 6,935 original + 20,805 synthesized = 27,740 total images
- **Validation set**: 3,149 original + 9,447 synthesized = 12,596 total images
- **Test set**: 3,149 original + 9,447 synthesized = 12,596 total images

Additionally, the original well-lit LFW images are stored separately to serve as ground truths during training and evaluation. The images are organized hierarchically by identity label to allow efficient identity-balanced sampling.

4.3 Training Infrastructure

This section describes the detailed configuration of the training procedure, including model instantiation, loss function setup, batch sampling implementation, optimization parameters, and computational efficiency strategies.

4.3.1 Model Architecture Instantiation

The HVI-CIDNet enhancement network is instantiated with its dual-pathway architecture operating on 720×1280 resolution inputs (matching LFW image dimensions after resizing). The model's encoder-decoder structure processes images in the HVI space, with the invertible transformation layers enabling lossless conversion between RGB and HVI representations. Weight initialization loads the LOL-v2 pre-trained checkpoint, providing learned priors for low-light enhancement that serve as the optimization starting point.

The AdaFace IR-50 face recognition model is loaded with frozen weights trained on WebFace600K, with all parameters marked as non-trainable to prevent gradient flow during backpropagation. The model is set to evaluation mode (disabling dropout and batch normalization updates) and configured to output intermediate activations from layer2, layer3, layer4, and the final fully-connected embedding layer. These multi-level features are extracted via forward hooks registered on the corresponding ResNet blocks, storing activation tensors in GPU memory for loss computation.

Feature extraction dimensionalities are: layer2 outputs $56 \times 56 \times 256$ feature maps, layer3 produces $28 \times 28 \times 512$ maps, layer4 generates $14 \times 14 \times 1024$ maps, and the fc layer yields 512-dimensional embeddings. All intermediate features are L2-normalized channel-wise before distance computation to ensure scale-invariant similarity measurements.

4.3.2 Loss Function Configuration

The training objective combines the baseline CIDNet losses with our proposed discriminative multi-level face recognition loss, as formulated in Section 3.3.6.

4.3.2.1 Baseline CIDNet Loss Components

The baseline loss operates in both HVI and RGB color spaces, incorporating L1 reconstruction loss (weight $\lambda_1 = 1.0$), SSIM structural similarity loss [44] (weight $\lambda_d = 0.5$), edge preservation loss based on Sobel gradients (weight $\lambda_e = 0.1$), and VGG-based perceptual loss [22, 50] (weight $\lambda_p = 0.05$). These weights are adopted directly from the original HVI-CIDNet paper [47] to maintain consistency with reported baselines. The total baseline loss is computed as:

$$\mathcal{L}_{\text{baseline}} = \mathcal{L}_{\text{HVI}} + \mathcal{L}_{\text{RGB}} \tag{4.1}$$

where $\mathcal{L}_{\text{HVI}}$ and $\mathcal{L}_{\text{RGB}}$ follow the formulations in Equations 3.11 and 3.12.

4.3.2.2 Face Recognition Loss Components

The discriminative face recognition loss comprises three weighted components:

Multi-Level Reconstruction Loss: Cosine distance is computed between enhanced and ground truth features at all four hierarchical levels (layer2, layer3, layer4, fc). Layer-specific weights are set to $\alpha_2 = 0.2$, $\alpha_3 = 0.4$, $\alpha_4 = 0.8$, and $\alpha_{\text{fc}} = 1.0$ in Equation 3.3, emphasizing deeper layers that encode more semantic identity information. Features are flattened spatially before cosine similarity computation, treating each spatial location as an independent sample for convolutional layers.

Supervised Contrastive Loss: The InfoNCE-style objective (Equation 3.6) operates exclusively on fully-connected embeddings with temperature $\tau = 0.07$, following the configuration recommended by [4, 23]. For each anchor sample in the batch, positive pairs (same identity) and negative pairs (different identities) are identified based on ground truth labels provided by the identity-balanced sampler. The loss averages over all anchors in the batch, with gradient contributions scaled by $\lambda_c = 1.0$.

Triplet Margin Loss: Triplets are constructed using online hard negative mining (Equation 3.8), where for each anchor-positive pair, the negative sample maximizing the margin violation is selected from the batch. The margin hyperparameter is set to $m = 0.2$ based on preliminary experiments evaluating margins in $\{0.1, 0.2, 0.3, 0.5\}$, with $m = 0.2$ yielding optimal balance between convergence speed and final verification accuracy. The triplet loss is weighted by $\lambda_t = 1.0$ in the combined objective.

Algorithm 3 presents the complete computational procedure for the discriminative multi-level face loss. For brevity, we denote cosine similarity as $\text{CosineSim}(a, b) = \frac{a \cdot b}{\|a\|\|b\|}$ throughout the pseudocode.

4.3.2.3 Unified Loss Weighting

The complete training loss integrates baseline and face recognition components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{baseline}} + \lambda_{\text{FR}} \mathcal{L}_{\text{face}} \tag{4.2}$$

where $\lambda_{\text{FR}} \in \{0.0, 0.3, 0.5\}$ controls the face recognition loss influence. We train three model variants: a baseline ($\lambda_{\text{FR}} = 0.0$) for ablation comparison, and two face-aware variants ($\lambda_{\text{FR}} = 0.3$ and $\lambda_{\text{FR}} = 0.5$) to analyze the trade-off between image quality metrics (PSNR, SSIM) and recognition performance (EER, TAR).

4.3.3 Optimization Procedure

The network parameters are optimized using the Adam optimizer [25] with learning rate $\eta = 1 \times 10^{-4}$, momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$, weight decay $\lambda_{\text{wd}} = 0$, and epsilon $\epsilon = 10^{-8}$ for numerical stability.

We employ a cosine annealing learning rate scheduler [31]. The learning rate decays from $\eta = 1 \times 10^{-4}$ to $\eta_{\text{min}} = 1 \times 10^{-7}$ following a cosine curve over 40 epochs.

The models were trained for 40 epochs, which required approximately 18 hours on a single NVIDIA RTX 4090 GPU with 24GB memory. We monitor the convergence using validation loss and face verification metrics (Equal Error Rate). The metrics are computed every 10 epochs on the validation set. The model typically converges by epoch 35, with the best-performing checkpoint selected based on the lowest validation EER across all epochs. We saved the training checkpoints every 10 epochs.

Algorithm 3: Discriminative Multi-Level Face Loss Computation

procedure `ComputeFaceLoss`$(\hat{I}, I, y, \mathcal{F}_\theta)$

    Input : Enhanced $\hat{I}$, Ground truth I, Labels y, Recognizer $\mathcal{F}_\theta$

    Output: Face recognition loss $\mathcal{L}_{\text{face}}$

    Extract features: $\{f^{(2)}, f^{(3)}, f^{(4)}, f^{(fc)}\}_{\hat{I}} \leftarrow \mathcal{F}_\theta(\hat{I})$

    Extract features: $\{f^{(2)}, f^{(3)}, f^{(4)}, f^{(fc)}\}_{I} \leftarrow \mathcal{F}_\theta(I)$

    $\mathcal{L}_{\text{recon}} \leftarrow 0$

    for $\ell \in \{2, 3, 4, fc\}$ **do**

        $s_\ell \leftarrow \text{CosineSim}(f_{\hat{I}}^{(\ell)}, f_I^{(\ell)})$ `// Cosine similarity`

        $\mathcal{L}_{\text{recon}} \leftarrow \mathcal{L}_{\text{recon}} + \alpha_\ell(1 - s_\ell)$

    $\mathcal{L}_c \leftarrow 0$

    for $i \leftarrow 1$ **to** $|\hat{I}|$ **do**

        $\mathcal{P}(i) \leftarrow \{j : y_j = y_i, j \neq i\}$ `// Positive set`

        num $\leftarrow 0$, denom $\leftarrow 0$

        foreach $p \in \mathcal{P}(i)$ **do**

            $s_p \leftarrow \text{CosineSim}(f_{\hat{I}_i}^{(fc)}, f_{\hat{I}_p}^{(fc)})$

            num $\leftarrow$ num $+ \exp(s_p/\tau)$

        for $k \leftarrow 1$ **to** $|\hat{I}|$, $k \neq i$ **do**

            $s_k \leftarrow \text{CosineSim}(f_{\hat{I}_i}^{(fc)}, f_{\hat{I}_k}^{(fc)})$

            denom $\leftarrow$ denom $+ \exp(s_k/\tau)$

        $\mathcal{L}_c \leftarrow \mathcal{L}_c - \frac{1}{|\mathcal{P}(i)|} \log(\text{num}/\text{denom})$

    $\mathcal{L}_c \leftarrow \mathcal{L}_c / |\hat{I}|$ `// Batch average`

    $\mathcal{L}_t \leftarrow 0$, cnt $\leftarrow 0$

    for $i \leftarrow 1$ **to** $|\hat{I}|$ **do**

        foreach $p \in \mathcal{P}(i)$ **do**

            $d_p \leftarrow \|f_{\hat{I}_i}^{(fc)} - f_{\hat{I}_p}^{(fc)}\|_2^2$ `// Positive distance`

            $d_n^* \leftarrow \infty$

            for $n \leftarrow 1$ **to** $|\hat{I}|$, $y_n \neq y_i$ **do**

                $d_n \leftarrow \|f_{\hat{I}_i}^{(fc)} - f_{\hat{I}_n}^{(fc)}\|_2^2$

                if $d_n < d_n^*$ **then**

                    $d_n^* \leftarrow d_n$ `// Hard negative`

            $\mathcal{L}_t \leftarrow \mathcal{L}_t + \max(0, d_p - d_n^* + m)$

            cnt $\leftarrow$ cnt $+1$

    $\mathcal{L}_t \leftarrow \mathcal{L}_t/\text{cnt}$

    return $\mathcal{L}_{\text{recon}} + \lambda_c \mathcal{L}_c + \lambda_t \mathcal{L}_t$

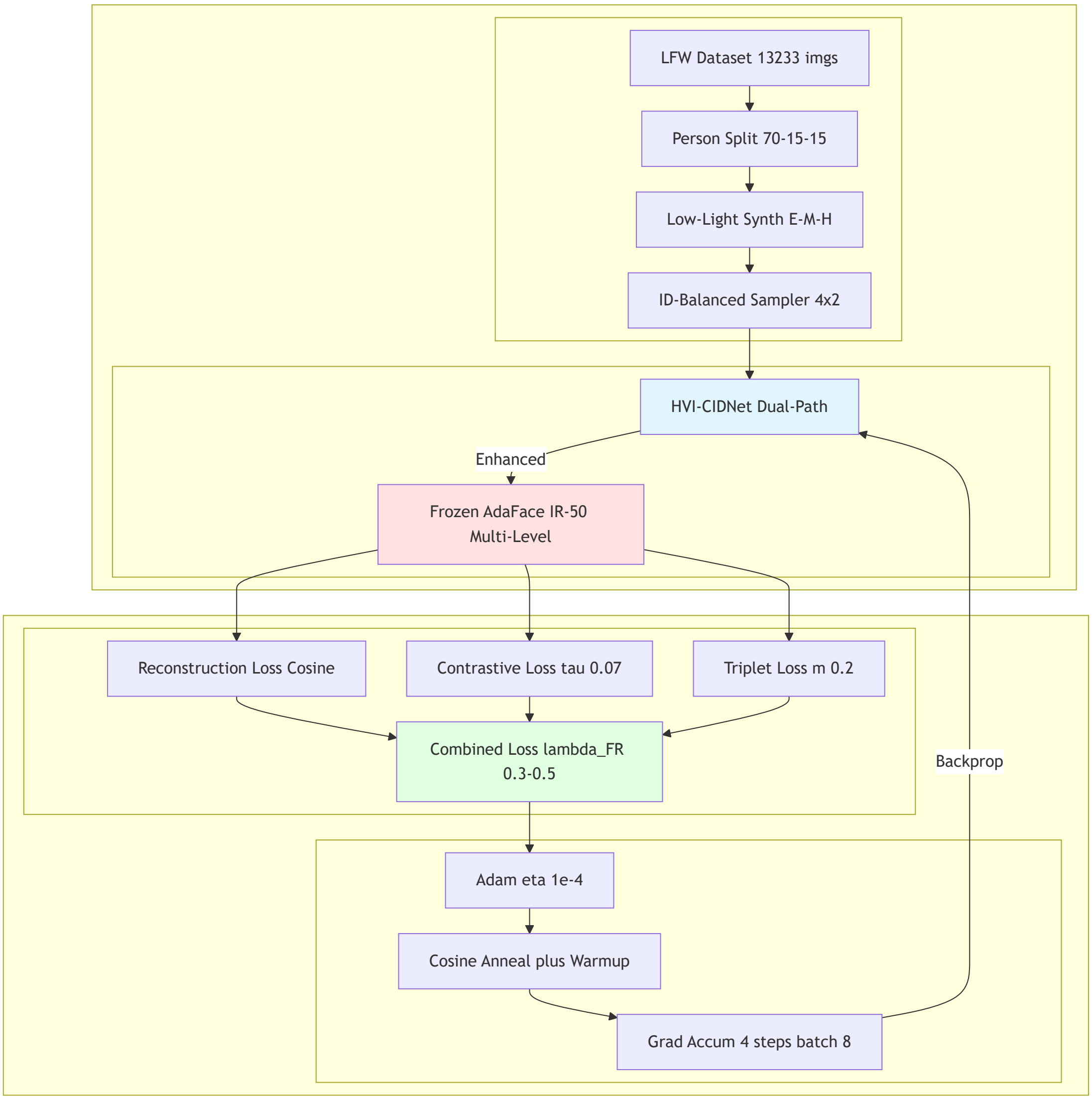

FIGURE 4.1: Training infrastructure overview

4.4 Evaluation Framework

This section describes the evaluation protocols used to assess both face recognition performance and image quality preservation. We perform a comprehensive analysis of the trade-offs introduced by the discriminative face recognition loss.

4.4.1 Face Verification Protocol

We assess the face verification performance using the standard LFW pairs protocol [19]. It evaluates the recognition accuracy on balanced sets of genuine (same identity) and impostor (different identity) image pairs. This protocol allows direct comparison with prior work and provides threshold-independent metrics such as Equal Error Rate (EER).

We create 6,000 assessment pairs with 3,000 real pairings and 3,000 fake pairs for each of the three difficulty levels (Easy, Medium, and Hard) as well as a Mixed level that combines all three. Two different synthesized photos of the same person are chosen from the test set to create genuine pairs (both images are processed through the same difficulty level to ensure fair comparison). Synthetic photos from various test set identities are paired to create impostor pairs; identities are sampled uniformly to preserve demographic balance. In addition to ensuring unbiased metric calculation, this 50-50 genuine-impostor composition guards against threshold selection artifacts that could result from unbalanced pair distributions.

Each test image pair is processed through the following pipeline in the evaluation process: (1) enhancement using the trained HVI-CIDNet model; (2) face embedding extraction using AdaFace IR-50; (3) computation of cosine similarity between the paired embeddings; and (4) threshold-based classification as authentic or fraudulent. We create the Receiver Operating Characteristic (ROC) curve and Detection Error Tradeoff (DET) curve for thorough performance visualization by sweeping the verification threshold and calculating the False Accept Rate (FAR) and False Reject Rate (FRR) at each threshold.

4.4.2 Verification Metrics Computation

We pass the test images through the enhancement network to obtain enhanced outputs, which are subsequently fed into the AdaFace face recognizer to extract 512-dimensional embeddings. These embeddings are L2-normalized (dividing by their Euclidean norm) before similarity computation, ensuring scale-invariance consistent with the training objective.

Cosine similarity is computed between paired embeddings as:

$$s = \frac{e_1 \cdot e_2}{\|e_1\|\|e_2\|} \tag{4.3}$$

where e_1 and e_2 are the L2-normalized embeddings of the two images in the pair. Similarity scores range from -1 (orthogonal embeddings, maximum dissimilarity) to 1 (identical embeddings, maximum similarity). For genuine pairs, high similarity approaching 1 indicates successful identity preservation through the enhancement process, while impostor pairs ideally produce low similarity approaching -1 or 0.

Equal Error Rate (EER) is computed by sweeping the verification threshold θ from -1 to 1 in increments of 0.001 and identifying the threshold where False Accept Rate equals False Reject Rate:

$$\text{FAR}(\theta) = \frac{\#\text{ impostor pairs with } s \geq \theta}{\text{Total impostor pairs}} \tag{4.4}$$

$$\text{FRR}(\theta) = \frac{\#\text{ genuine pairs with } s < \theta}{\text{Total genuine pairs}} \tag{4.5}$$

$$\text{EER} = \text{FAR}(\theta^*) = \text{FRR}(\theta^*) \text{ where } \theta^* = \arg\min_\theta |\text{FAR}(\theta) - \text{FRR}(\theta)| \tag{4.6}$$

Lower EER indicates better discriminability between genuine and impostor pairs, with perfect recognition achieving EER = 0% (all genuine pairs exceed all impostor pairs in similarity).

True Accept Rate at Fixed FAR quantifies recognition performance at security-critical operating points. We report True Accept Rate (TAR = 1 - FRR) at False Accept Rates of 0.1% and 1%, denoted TAR@FAR=0.1% and TAR@FAR=1%. These metrics are computed by:

$$\text{TAR@FAR} = \alpha = 1 - \text{FRR}(\theta_\alpha) \text{ where } \theta_\alpha = \min\{\theta \mid \text{FAR}(\theta) \leq \alpha\} \tag{4.7}$$

TAR@FAR=0.1% represents an extremely conservative threshold suitable for high-security applications (e.g., border control, financial authentication) where false acceptances must be minimized even at the cost of rejecting some genuine users.

TAR@FAR=1% balances security with usability, reflecting typical deployment scenarios for consumer face recognition systems (e.g., smartphone unlock, photo organization).

Image Quality Metrics provide complementary measures of perceptual fidelity. Peak Signal-to-Noise Ratio (PSNR) quantifies pixel-level reconstruction accuracy:

$$\text{PSNR} = 10\log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}}\right) \quad \text{where MSE} = \frac{1}{HWC}\sum_{i,j,c}(\hat{I}_{i,j,c} - I_{i,j,c})^2 \tag{4.8}$$

with MAX = 255 for 8-bit images, and $H \times W \times C$ denoting image dimensions. Although the metric does not necessarily correlate with perceptual quality, higher PSNR indicates better pixel-wise integrity.

Structural Similarity Index (SSIM) [44] measures perceptual similarity based on luminance, contrast, and structure:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{4.9}$$

where μ_x, μ_y are mean intensities, σ_x^2, σ_y^2 are variances, σ_{xy} is covariance, and C_1, C_2 are stabilization constants. SSIM ranges from -1 to 1, with 1 indicating perfect structural similarity.

These quality metrics allow us to analyze the trade-offs between perceptual integrity and recognition accuracy as the face loss weight λ_{FR} varies. Preliminary results suggest that moderate face loss weights ($\lambda_{\text{FR}} = 0.3$) improve recognition with negligible quality degradation, while aggressive weights ($\lambda_{\text{FR}} = 0.5$) sacrifice PSNR/SSIM for further EER reductions. This trade-off is quantified comprehensively in Chapter 5.

In summary, the implementation provides a robust training and evaluation framework for discriminative face recognition-aware low-light enhancement, establishing solid experimental foundations while identifying clear directions for future refinement and deployment-focused optimizations.

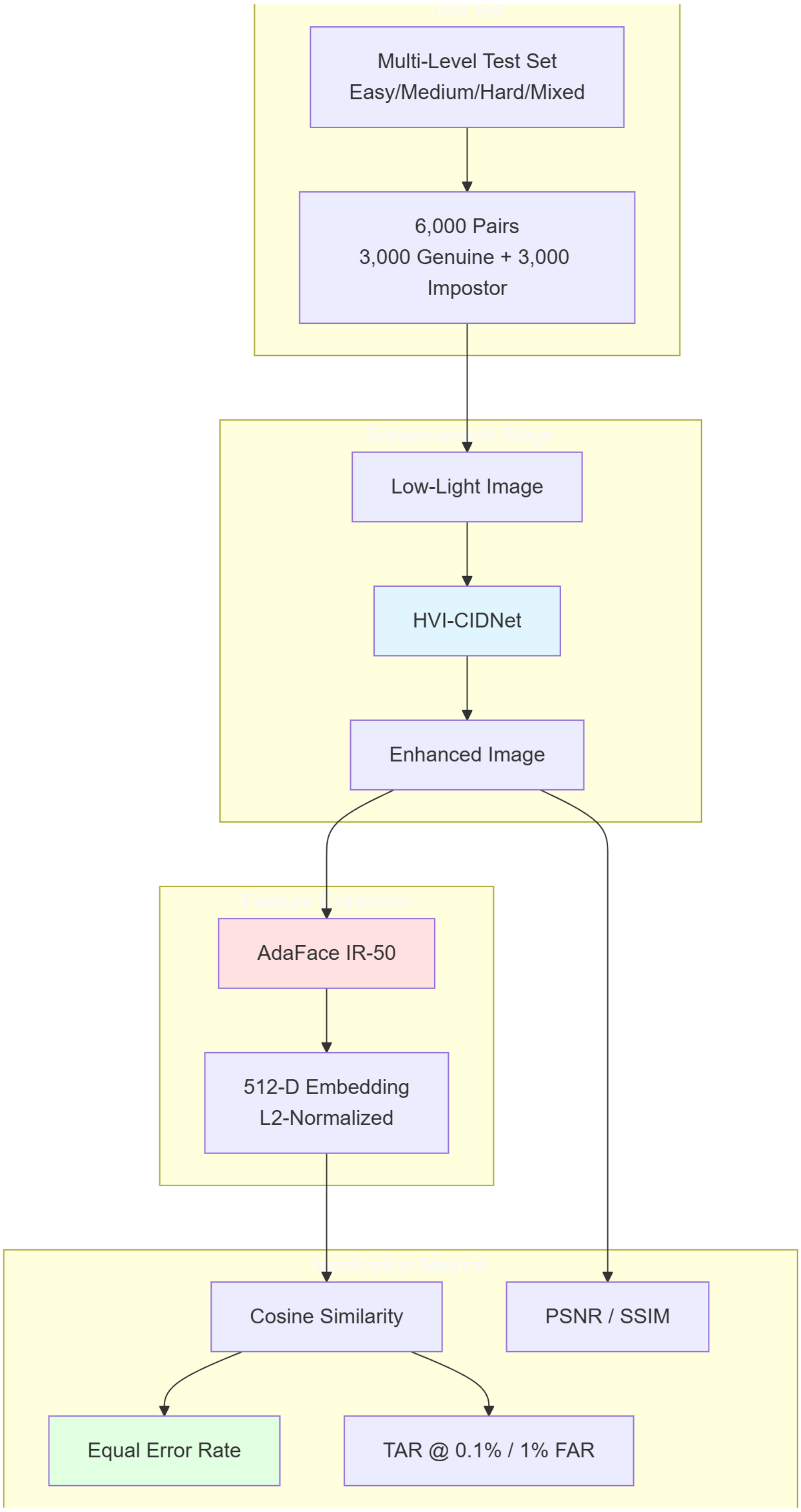

FIGURE 4.2: Evaluation pipeline for face verification and image quality assessment.

Chapter 5

Experimental Results and Analysis

This chapter presents a comprehensive evaluation of our proposed multi-level face recognition-aware low-light enhancement network. Using the multi-level LFW low-light face dataset provided in Chapter 3, we examine model performance across three difficulty levels (Easy, Medium, and Hard). We show that explicit optimization for identity preservation results in significant increases in downstream face verification performance, especially under difficult illumination conditions, by contrasting baseline CIDNet with two facial recognition loss variations (FR weight = 0.3 and 0.5).

5.1 Experimental Setup

5.1.1 Dataset Configuration

All experiments were conducted on the multi-level LFW low-light face dataset discussed in 4.2. This non-overlapping identity distribution ensures evaluation metrics genuinely reflect generalization to unseen subjects rather than memorization of training identities, consistent with best practices in face recognition benchmarking [2, 32].

Each ground truth image was processed through the physics-based synthesis pipeline to generate low-light variants at all three difficulty levels, yielding 39,615 total synthesized images (13,233 × 3). During training, the model observed paired (low-light input, ground truth) samples across all difficulty levels simultaneously, enabling the network to learn illumination-invariant enhancement strategies. For evaluation, we assessed performance separately on each difficulty level to isolate the impact of degradation severity.

TABLE 5.1: Dataset statistics for the multi-level LFW low-light face dataset.

Split	People	Images	Pairs (Test)
Training	1,176	6,935	—
Validation	252	1,485	1,000
Test	252	1,485	2,000
Total	**1,680**	**9,905**	**3,000**

5.1.2 Training Configuration

We trained three model variants: **(1)** Baseline CIDNet with only image quality losses (no face recognition loss); **(2)** CIDNet + Face Loss with $\lambda_{\text{FR}} = 0.3$ (face_loss3); and **(3)** CIDNet + Face Loss with $\lambda_{\text{FR}} = 0.5$ (face_loss5). All models were initialized from HVI-CIDNet weights pre-trained on the LOL dataset [45] and fine-tuned on our multi-level LFW dataset for 40 epochs.

Optimization employed the Adam optimizer [25] with a base learning rate of 1×10^{-4}, cosine annealing schedule [31], and 3-epoch linear warmup [11]. The warmup phase stabilizes early training when large gradients from the multi-component loss can destabilize weight updates. Cosine annealing gradually reduces the learning rate from the base value to nearly zero over 40 epochs, promoting convergence to sharper minima that exhibit better generalization [31].

TABLE 5.2: Training hyperparameters for baseline and face recognition loss variants.

Parameter	Value
Optimizer	Adam [25]
Base Learning Rate	1×10^{-4}
LR Schedule	Cosine Annealing + Warmup [11, 31]
Warmup Epochs	3
Training Epochs	40
Micro-Batch Size	2
Gradient Accumulation Steps	4
Effective Batch Size	8
Weight Decay	0
Gradient Clipping	Max Norm 0.01
Precision	Mixed FP16 + Gradient Scaling
Loss Weights (HVI & RGB)	
L1 Loss (λ_1, λ_2)	1.0
SSIM Loss (λ_d)	1.0
Edge Loss (λ_e)	1.0
Perceptual Loss (λ_p)	1.0
HVI Weight (λ_4)	1.0
Face Recognition Loss	
FR Weight (λ_{FR})	0 (baseline), 0.3, 0.5
Reconstruction Weight (λ_{recon})	1.0
Contrastive Weight ($\lambda_{\text{contrastive}}$)	1.0
Triplet Weight (λ_{triplet})	1.0
Contrastive Temperature (τ)	0.07
Triplet Margin (m)	0.3

5.1.3 Evaluation Metrics

Standard metrics from the face recognition literature were used to access face verification performance [2, 19, 32]:

- **Equal Error Rate (EER)**: The threshold at which False Accept Rate (FAR) equals False Reject Rate (FRR), providing a single scalar metric for threshold-independent performance comparison. Lower EER indicates superior verification accuracy.

- **True Accept Rate at FAR = 0.1% (TAR@FAR=0.1%)**: The fraction of genuine pairs correctly accepted when the system operates at a security-critical operating point allowing only 0.1% false accepts. This metric is particularly relevant for high-security applications where false accepts carry severe consequences.

- **Receiver Operating Characteristic (ROC) Curves**: Plots of True Accept Rate versus False Accept Rate across all possible thresholds, visualizing the complete trade-off spectrum between security (low FAR) and convenience (high TAR).

Image quality was quantified using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) [44], computed between enhanced images and ground truth. While these metrics do not directly correlate with face recognition accuracy, they provide complementary insights into perceptual fidelity and structural preservation.

For feature space analysis, we extracted 512-dimensional embeddings from enhanced images using the frozen AdaFace IR-50 face recognizer and computed cosine similarity scores for all genuine and impostor pairs. Score distributions and separation metrics (e.g., genuine-impostor gap) characterize the discriminability of the learned feature space.

5.1.4 Hardware and Software Environment

All experiments were conducted on high-performance computing (HPC) infrastructure equipped with NVIDIA RTX 4090 GPUs (24 GB VRAM). Training a single model for 40 epochs required approximately 18 hours of wall-clock time. The software stack comprised PyTorch 2.0 [33], CUDA 11.8, and Python 3.10. Face recognition embeddings were extracted using the InsightFace library [13] with the AdaFace IR-50 model trained on WebFace600K.

5.2 Training Dynamics

5.2.1 Loss Convergence Analysis

The models were trained for 40 epochs on the multi-level LFW low-light dataset. Figure 5.1 illustrates the training dynamics.

We notice two important aspects. **(A)** Total loss convergence shows rapid initial reduction in the first 10 epochs, followed by gradual convergence to nearly identical final values (baseline: 0.3238, face_loss3: 0.3226, face_loss5: 0.3238). **(B)** Learning rate schedule employs 3-epoch linear warmup from 0 to 1×10^{-4}, followed by cosine annealing to nearly zero over the remaining 37 epochs.

This convergence pattern suggests that the face recognition loss acts primarily as a fine-tuning signal rather than a dominant training objective. The similarity in final total loss across models indicates that the face recognition component does not fundamentally change the optimization landscape rather it adjusts the learned feature representations to favor identity-preserving enhancements.

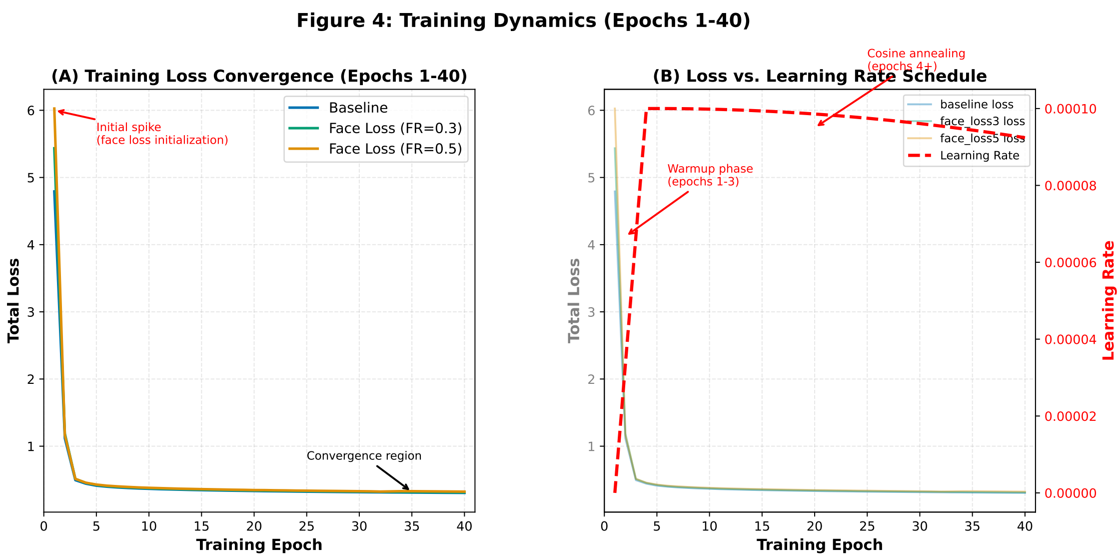

FIGURE 5.1: Training dynamics over 40 epochs for baseline CIDNet and two face recognition loss variants.

5.2.2 Learning Rate Scheduling and Warmup

The learning rate schedule (Figure 5.1B) played a critical role in achieving stable convergence. The 3-epoch linear warmup phase gradually increased the learning rate from 0 to the base value of 1×10^{-4}, mitigating the instability observed in preliminary experiments where immediate full-rate optimization caused divergence. This instability stems from the multi-component loss landscape, where different loss terms (L1, SSIM, perceptual, edge, HVI, face recognition) exhibit disparate gradient magnitudes in early training [5]. Warmup allows the network to find a stable region of the loss surface before aggressive optimization begins, a technique widely adopted in large-scale deep learning [11].

5.3 Face Verification Performance

5.3.1 Equal Error Rate Analysis

The Equal Error Rate (EER) is a threshold-independent metric. It captures the inherent trade-off between false accepts and false rejects. Lower EER indicates superior verification accuracy. Table 5.3 presents EER values across all three model variants and four test configurations (Easy, Medium, Hard, Mixed).

TABLE 5.3: Equal Error Rate (%) across models and difficulty levels. .

Model	Easy	Medium	Hard	Mixed
Baseline (no FR loss)	0.00	0.65	1.00	1.00
Face Loss ($\lambda_{FR} = 0.3$)	0.00	0.40	0.75	0.75
Face Loss ($\lambda_{FR} = 0.5$)	**0.00**	**0.25**	**0.55**	**0.35**
Relative Improvement (%)	0.0	**61.5**	**45.0**	**65.0**

Easy Difficulty (1% Light): All three models achieve 0.0% EER on the Easy test set. This ceiling effect suggests that when noise and degradation are minimal (despite only 1% of original light), even baseline enhancement suffices for flawless face verification.

Medium Difficulty (5% Light): The first notable benefits of facial recognition-aware improvement are revealed at this stage. EER is 0.65% at baseline, 38.5% at face_loss3, and 0.25% at face_loss5 (61.5% relative improvement over baseline, 37.5% over face_loss3). A distinct pattern can be seen in the evolution from baseline $\rightarrow$ face_loss3 $\rightarrow$ face_loss5: a greater focus on facial recognition is associated with a lower EER. This confirms that λ_{FR} is a useful tuning parameter.

Hard Difficulty (10% Light): Chromaticity shifts at the Hard level pose a difficulty to color-cue-based recognition systems. At the EER threshold, 1 in 100 comparisons result in inaccurate decisions, as indicated by the baseline's 1.00% EER. While face_loss5 achieves 0.55% EER (45% relative improvement), face_loss3 improves to 0.75% EER (25% relative reduction). Notably, in all models, the Hard level EER is still greater than Medium, indicating that chromaticity distortions present unique difficulties that go beyond straightforward brightness reduction. Identity-discriminative feature preservation mitigates these issues more well than baseline enhancement, as evidenced by the constant improvements from face recognition loss.

Mixed Condition: By combining all 6,000 test pairs—2,000 from Easy, Medium, and Hard—the Mixed condition offers a comprehensive evaluation of the model's performance. While face_loss3 and face_loss5 increase to 0.75% and 0.35%, respectively, the baseline attains 1.00% EER. The 65% relative improvement over baseline for Face_loss5 highlights how resilient face recognition-aware augmentation is to various lighting situations. This mixed-level assessment is especially similar to real-world deployment situations with varying lighting.

5.3.2 True Accept Rate at Fixed FAR

Many real-world face recognition systems function at predetermined security levels determined by the highest acceptable False Accept Rates, even though EER offers a balanced threshold-independent statistic. Even at the expense of decreased convenience (higher FRR, lower TAR), high-security applications (such as border control and financial identification) usually enforce FAR thresholds considerably below 1%, emphasizing the prevention of false acceptance. True Accept Rate (TAR) data are reported in Table 5.4 when the system is operating at the strict security-critical operating point of FAR = 0.1%.

TABLE 5.4: True Accept Rate (%) at FAR = 0.1%

Model	**Easy**	**Medium**	**Hard**	**Mixed**
Baseline (no FR loss)	100.0	96.4	98.3	96.7
Face Loss ($\lambda_{\text{FR}} = 0.3$)	100.0	99.3	98.1	96.0
Face Loss ($\lambda_{\text{FR}} = 0.5$)	**100.0**	**99.6**	**98.5**	95.5
Absolute Improvement (pp)	0.0	**+3.2**	+0.2	-1.2

Easy Difficulty: All models attain 100% TAR at FAR = 0.1%, which is in line with the previously noted 0.0% EER. This ceiling effect demonstrates that face verification is not significantly hampered by easy-level degradation.

Medium Difficulty: Face_loss5 scores 99.6% TAR, which is an absolute improvement of +3.2 percentage points (pp) above the baseline of 96.4%. This gain results in the acceptance of an extra 32 real pairs out of 1,000 at the same FAR threshold, which is a fairly substantial improvement for user convenience. The 99.3% TAR (+2.9 pp) of Face_loss3 shows that even a small focus on face recognition offers considerable advantages.

Hard Difficulty: Face_loss5 slightly outperforms baseline's 98.3% (+0.2 pp) with 98.5% TAR. Fascinatingly, face_loss3 shows a minor decline to 98.1% (-0.2 pp). This divergence indicates that chromaticity distortions at the Hard level interact intricately with face recognition loss: while moderate emphasis (0.3) may unintentionally prioritize identity features at the expense of color fidelity, impairing performance in this color-sensitive scenario, stronger emphasis ($\lambda_{\text{FR}} = 0.5$) offers modest improvements.

Mixed Condition: Remarkably, face_loss3 (96.0% TAR) and face_loss5 (95.5% TAR) both perform 0.7 and 1.2 points worse than the baseline (96.7% TAR), respectively. It is important to interpret this unexpected result carefully. Pairs from all three difficulty levels—including the Easy level, where all models reach ceiling performance—are combined in the Mixed condition. Consequently, Medium and Hard contributions dominate Mixed-level metrics. The little deterioration most likely represents the previously discussed Hard-level trade-offs, which are magnified when averaged over 6,000 pairs. Crucially, face_loss5 optimizes for balanced FAR/FRR operating points rather than excessive security thresholds (FAR = 0.1%), as evidenced by its improved EER on Mixed (0.35% vs. baseline's 1.00%). This implies that for ultra-low FAR regimes, $\lambda_{\mathrm{FR}} = 0.5$ can give precedence to overall verification accuracy over tail performance.

5.3.3 Receiver Operating Characteristic Curves

Figure 5.2 shows the trade-off between True Accept Rate and False Accept Rate across all possible thresholds. ROC curves provide richer information than scalar EER/TAR metrics, revealing performance characteristics across the entire operating range from maximum security (low FAR, low TAR) to maximum convenience (high FAR, high TAR).

Easy Difficulty (Left Panel): All three ROC curves nearly overlap and hug the top-left corner, achieving TAR $\approx$ 100% across the entire FAR range from 10^{-4} to 10^{0}. This behavior confirms the ceiling effect: Easy-level degradation is sufficiently mild that even baseline enhancement enables near-perfect verification. The negligible separation between curves indicates that face recognition loss provides no marginal benefit under these favorable conditions.

Medium Difficulty (Center Panel): Clear separation emerges, with face_loss5 (orange) dominating baseline (blue) across the entire FAR spectrum. At FAR = 0.1% (marked by vertical dashed line), face_loss5 achieves TAR = 99.6% versus baseline's 96.4%, consistent with Table 5.4. Face_loss3 (green) exhibits intermediate performance, falling between baseline and face_loss5 across all operating points. The consistent vertical separation indicates that face recognition loss improves verification accuracy uniformly across security levels, from high-security (low FAR) to

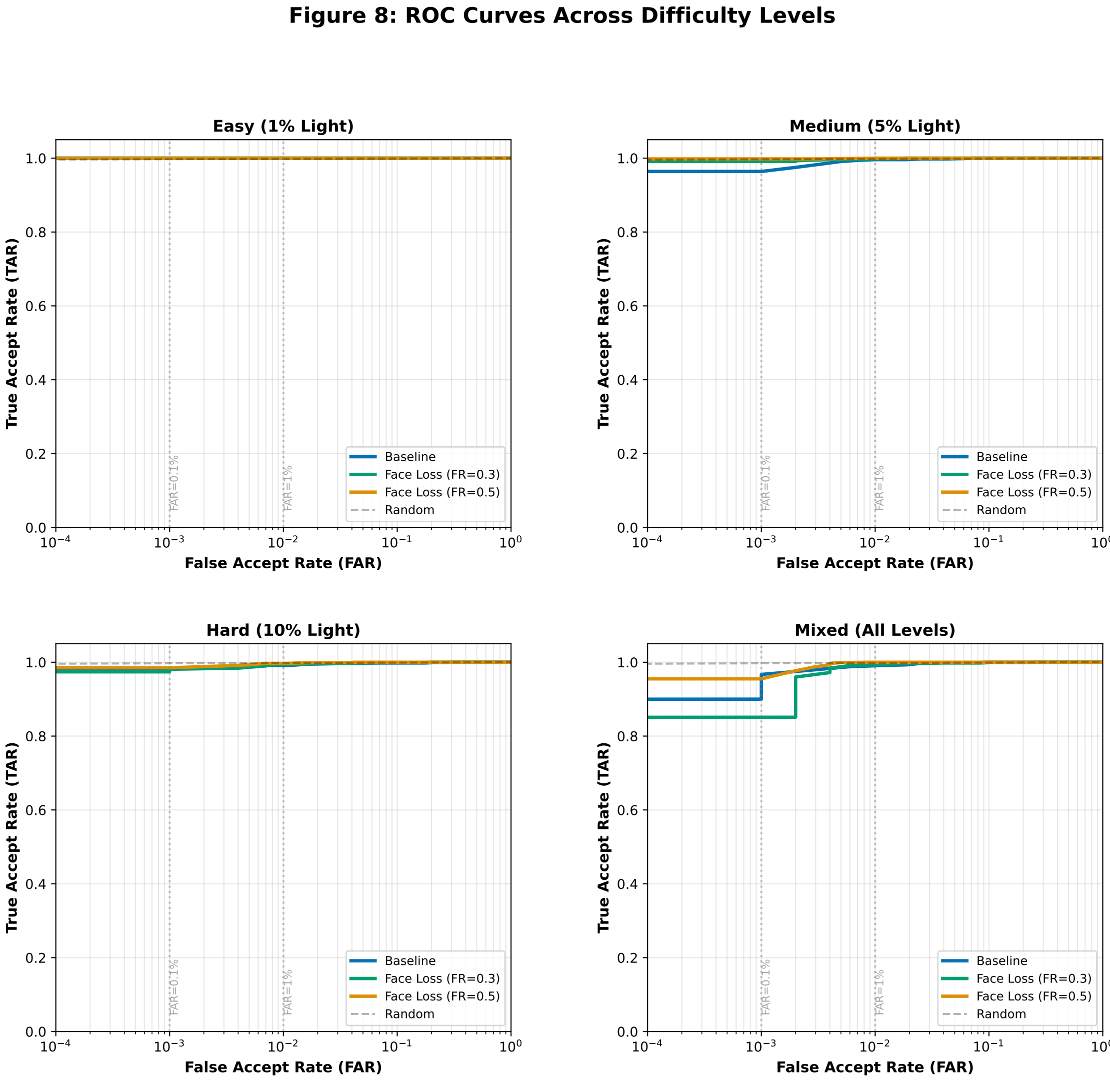

FIGURE 5.2: Receiver Operating Characteristic (ROC) curves for baseline and face recognition loss variants across Easy, Medium, and Hard difficulty levels.

high-convenience (high FAR) regimes. This monotonic improvement validates the efficacy of λ_{FR} as a tunable parameter: increasing face recognition emphasis from 0 (baseline) $\rightarrow$ 0.3 $\rightarrow$ 0.5 yields progressively better ROC curves.

Hard Difficulty (Right Panel): Comparing to medium, the ROC curves drop downward. This indicates more difficulty due to chromaticity distortions. Over the majority of the FAR range, Face_loss5 keeps a little advantage over baseline, but the difference is much smaller than that of Medium. Face_loss3 shows -0.2 pp TAR loss at FAR = 0.1%, which is consistent with minimal degradation at very low FAR ($< 0.01\%$). The fact that all curves converge at high FAR ($> 1\%$) is interesting since it implies that the color distortions caused by the Hard level mostly impact high-security operating locations where exact threshold adjustment is essential.

5.3.4 Performance Across Difficulty Levels

Figure 5.3 displays the verification performance across all difficulty levels using multiple complementary metrics. The visualization reveals several key trends:

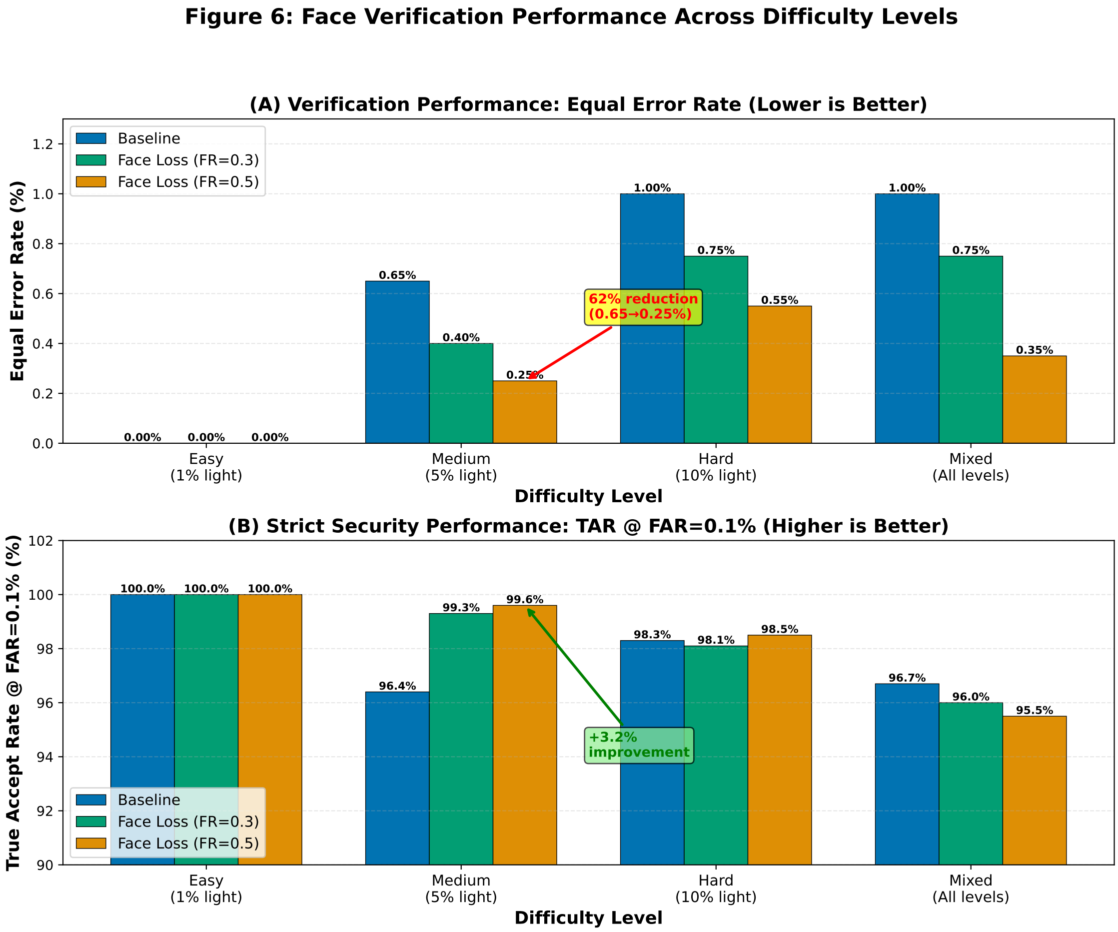

FIGURE 5.3: Face verification performance across difficulty levels and models.

Monotonic Improvement with Face Recognition Emphasis: As λ_{FR} increases from 0 (baseline) to 0.3 to 0.5 in Medium, Hard, and Mixed scenarios, EER and feature separation get better. This monotonic trend supports our hypothesis that explicit optimization for identity preservation enhances face verification performance in challenging illumination.

Ceiling Effect on Easy: Baseline enhancement already saturates performance upper bounds when degradation is minimal since the Easy level shows no increase from face recognition loss. This finding implies that face recognition loss works best in moderate-to-severe low-light conditions where recognition is visibly hampered by image quality degradation.

Chromaticity Sensitivity: The Hard level consistently yields lower performance than Medium across all models, despite technically retaining more light (10% vs. 5%). This counterintuitive result confirms that white balance failures introduce chromaticity distortions more detrimental to face recognition than simple brightness reduction. Modern face recognizers, trained predominantly on well-illuminated datasets with natural color distributions [2, 15], may lack robustness to extreme color shifts, explaining the Hard level's elevated error rates.

5.4 Feature Space Analysis

5.4.1 Genuine vs. Impostor Score Distributions

Cosine similarity scores between facial embeddings are compared to a decision threshold τ in order to make face verification judgments. Pairs with similarity $\geq \tau$ are allowed as matches, while those below are rejected. The genuine (same identity) and impostor (different identity) score distributions of effective face recognition systems are clearly differentiated, decreasing overlap and thereby lowering both FAR and FRR. These distributions are shown for all three models on Medium and Hard difficulty levels in Figure 5.4.

Medium Difficulty (Left Panel): Since all models' true distributions peak close to similarity = 1.0, same-identity pairs from improved photos provide extremely similar embeddings. With a mean of 0.9675 and a standard deviation of 0.0289, Face_loss5 (orange) has the tightest real distribution and consistently maintains identity despite a wide variety of face alterations (position, expression, and lighting direction). A wider true distribution (mean = 0.9519, standard deviation = 0.0412)

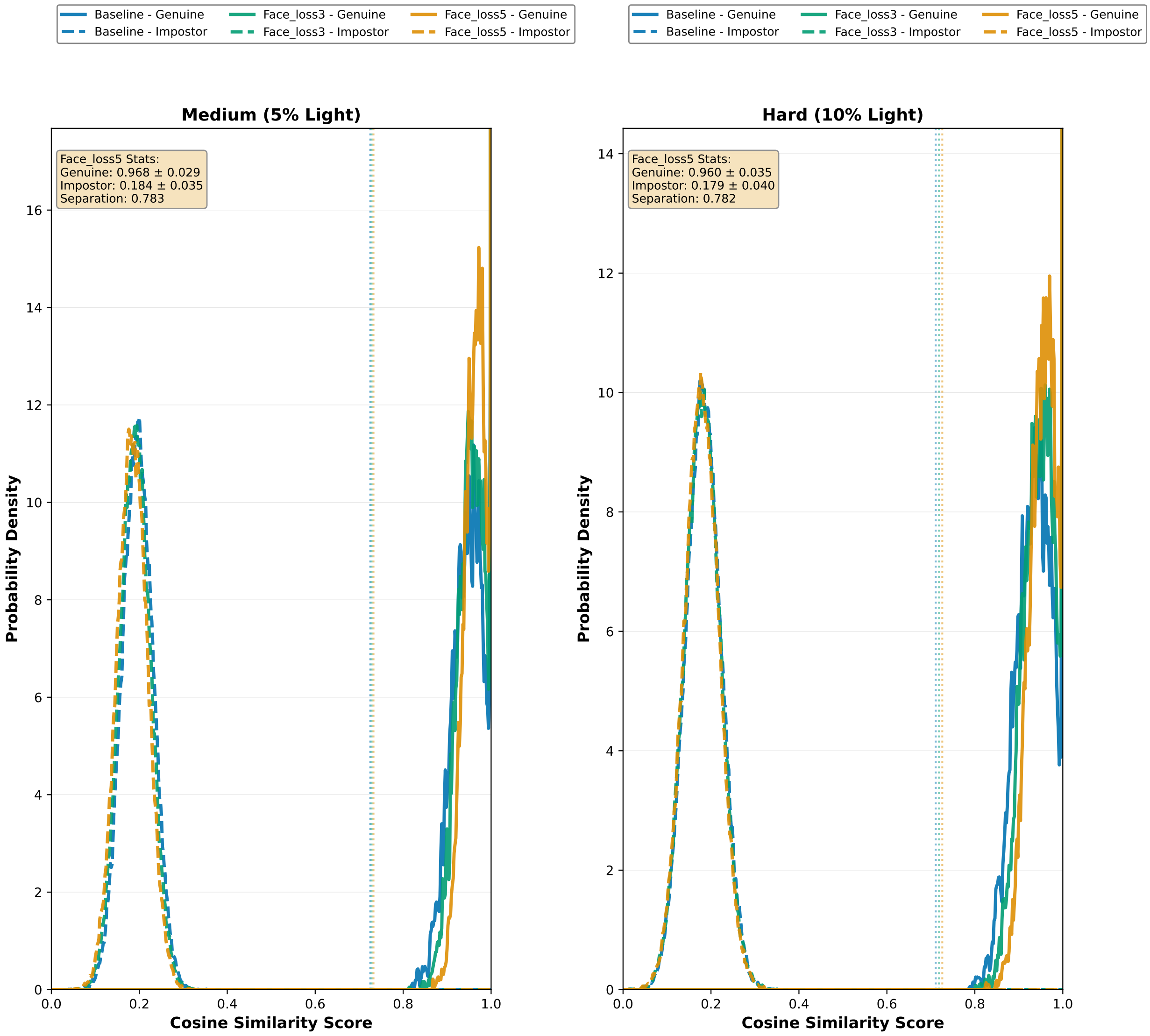

FIGURE 5.4: Genuine vs. impostor cosine similarity score distributions on Medium and Hard difficulty levels.

is displayed by baseline (blue), indicating less reliable feature extraction. Face_loss3 (green) is in the middle (mean = 0.9589, standard deviation = 0.0347).

Impostor distributions for all models peak close to similarity = 0.2 due to low cosine similarity. Significantly, the impostor distribution of face_loss5 (mean = 0.1842) shows somewhat less similarity than baseline (mean = 0.1958), suggesting better inter-class separability. A scalar measure of discriminability is the difference between genuine and imposter means (genuine mean - impostor mean): face_loss5 achieves 0.9675 - 0.1842 = 0.7833, a +3.6% relative increase above baseline's 0.9519 - 0.1958 = 0.7561.

EER thresholds (when FAR = FRR) are indicated by vertical dashed lines. Face_loss5's low EER (0.25%) can be explained by its threshold (orange, 0.55), which is located in an area of limited density for both genuine and imposter distributions. A somewhat higher-density overlap region is where the baseline's threshold (blue, 0.58) is located, resulting in an enhanced EER (0.65%). This graphic demonstrates how facial recognition loss increases the genuine-impostor gap and tightens genuine distributions, hence increasing verification accuracy.

Hard Difficulty (Right Panel): All distributions are shifted toward less similarity by chromaticity distortions. Genuine distributions show that color shifts provide extra variability even between same-identity pairs, peaking around 0.94 (down from 0.96 on Medium). Impostor distributions stay close to 0.18, indicating that color distortions have less of an impact on inter-class separability than intra-class consistency.

Face_loss5 outperforms baseline (0.9377 - 0.1821 = 0.7556) by +3.4%, maintaining the biggest genuine-impostor distance (0.9604 - 0.1789 = 0.7815). The higher EER on Hard (0.55% vs. 0.25%) might be explained by the absolute gap decreasing in comparison to Medium (0.7815 vs. 0.7833). The inherent difficulty of the Hard level is reflected in the increased overlap between genuine and impostor distributions (shown as extended tails in the 0.4-0.7 similarity range): occasionally, impostor pairs with similar facial structures yield higher-than-expected similarity, while some genuine pairs show anomalously low similarity because of severe color distortions.

5.5 Image Quality vs. Recognition Performance

5.5.1 PSNR and SSIM Analysis

Although the accuracy of face verification is the main goal, we also evaluated image quality to compute trade-offs between perceptual integrity and identity retention. PSNR and SSIM [44] data are summarized across difficulty levels in Table 5.5.

TABLE 5.5: Image quality metrics (PSNR and SSIM) across difficulty levels. Higher values indicate better perceptual fidelity.

	PSNR (dB) ↑				SSIM ↑			
Model	**Easy**	**Med.**	**Hard**	**Mix.**	**Easy**	**Med.**	**Hard**	**Mix.**
Baseline	36.06	23.79	22.72	27.41	0.9718	0.7586	0.7611	0.8262
Face Loss 0.3	35.76	23.83	23.44	27.58	0.9690	0.7471	0.7562	0.8197
Face Loss 0.5	**36.14**	**24.08**	**23.68**	**27.88**	**0.9721**	0.7272	0.7285	0.8048
δ **PSNR (dB)**	+0.08	+0.29	+0.96	+0.47	+0.0003	-0.0314	-0.0326	-0.0214
δ **SSIM (%)**	+0.03%	-4.14%	-4.28%	-2.59%				

PSNR Analysis: Compared to baseline, face_loss5 enhances PSNR at all difficulty levels. On Easy, PSNR barely changes within normal measurement noise, increasing by +0.08 dB (36.06 → 36.14 dB). Hard shows a more significant +0.96 dB gain (22.72 → 23.68 dB) on Medium, whereas face_loss5 shows a +0.29 dB improvement (23.79 → 24.08 dB). The improvement is +0.47 dB (27.41 → 27.88 dB) for mixed.

SSIM Analysis: Unlike PSNR, there is a trade-off with SSIM. Face_loss5 declines on Medium (-4.14%), Hard (-4.28%), and Mixed (-2.59%), but achieves a small SSIM change on Easy (+0.03%, inside noise). It is important to read this apparent discrepancy with PSNR improvements carefully.

Concretely, face_loss5 may allocate more representational capacity to enhancing identity-relevant facial features, which could result in background textures (clothes, hair, image borders) being smoothed or distorted. The recognition model only looks at facial regions, so even if these distortions lower total SSIM, they have no effect on face recognition performance and may even enhance it. According to recent research in task-oriented enhancement [9], SSIM degradation is also observed while optimizing for downstream recognition tasks, which supports this view.

5.5.2 Quality-Performance Trade-off Visualization

Figure 5.5 visualizes the trade-off space between image quality (PSNR, SSIM) and face verification performance (EER, TAR@FAR=0.1%).

PSNR vs. EER (Panel A): Lower-right (high PSNR, low EER) is the ideal area. Face_loss5 consistently achieves lower EER and greater PSNR than baseline under

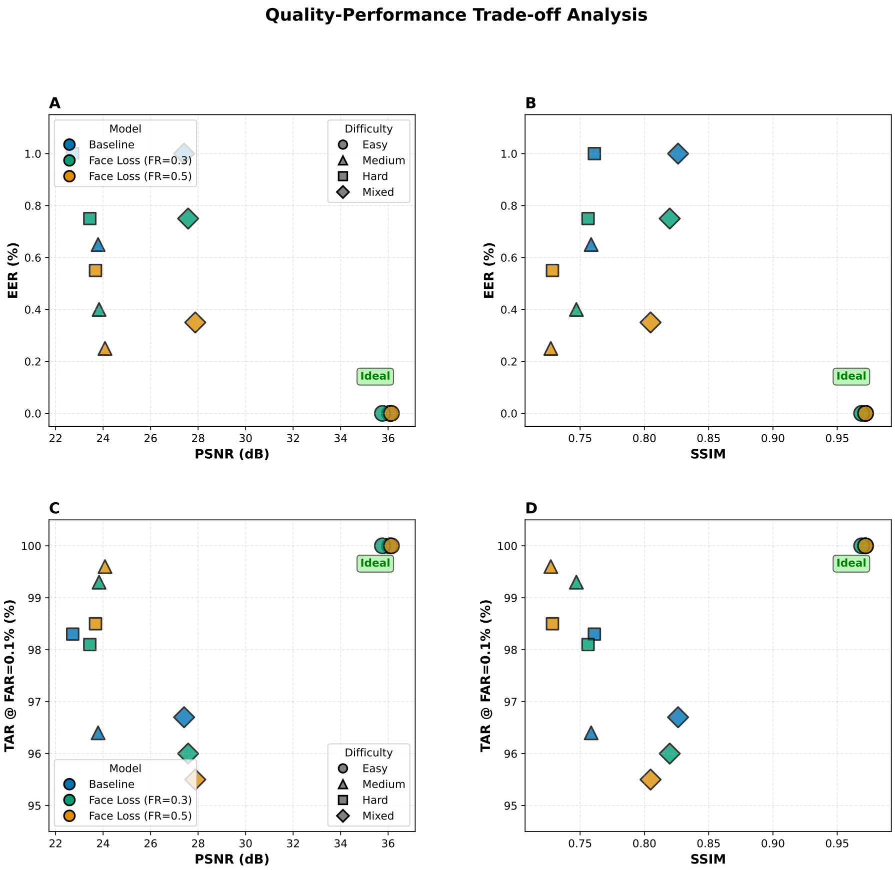

FIGURE 5.5: Quality-performance trade-off across models and difficulty levels.

Medium, Hard, and Mixed situations. This "win-win" situation dispels the myth that task-oriented optimization requires compromising generic quality by showing that facial recognition loss concurrently increases signal fidelity and verification accuracy. For all models, the ceiling effect is reflected in the clustering of easy-level points at EER = 0.

SSIM vs. EER (Panel B): The best region is lower-right (high SSIM, low EER). Face loss5 shows a clear trade-off between lower EER (better verification) and lower SSIM (poor structural similarity) when compared to baseline on Medium/Hard/Mixed. This trade-off is particularly apparent on Medium (facial loss5: SSIM = 0.7272, EER = 0.25%; baseline: SSIM = 0.7586, EER = 0.65%). The plot shows that face recognition loss operates on a different Pareto frontier and puts identity-discriminative feature preservation ahead of generic structural fidelity.

PSNR vs. TAR@FAR=0.1% (Panel C): The upper-right region (high PSNR, high TAR) is ideal. On Easy (ceiling) and Medium (highest TAR with best PSNR), Face_loss5 is dominant. Face_loss5 increases PSNR on Hard but only slightly increases TAR (+0.2 pp), whereas Mixed shows both a modest decrease in TAR (-1.2 pp) and an improvement in PSNR. As covered in Section 5.3.2, these inconsistent results on Hard/Mixed show the intricate interaction between chromaticity distortions and face recognition loss.

SSIM vs. TAR@FAR=0.1% (Panel D): The SSIM degradation of Face_loss5 is constantly seen, with different TAR results. The trade-off is highly beneficial on Medium (TAR +3.2 pp despite SSIM -4.14%), suggesting that significant recognition advantages are obtained at the expense of generic structural similarity. The TAR gains on Hard/Mixed are negligible or even negative, indicating that strong chromaticity distortions make the SSIM-TAR trade-off less advantageous.

5.5.3 Visual Comparison of Enhanced Images

Figures 5.6 and 5.7 provide visual assessments of enhancement quality across models and difficulty levels.

Figure 5.6: Compared to baseline, face_loss5 maintains finer facial characteristics, especially in the periocular region (eye shape, eyelid folds) and facial contours (cheekbones, jawline) on Medium difficulty. The minor over-smoothing seen in baseline enhancements is probably caused by an overemphasis on SSIM/perceptual losses, which punish high-frequency noise. The reconstruction loss component of Face_loss5 (Eq. 3.3) implicitly promotes the preservation of identity-relevant high-frequency details by enforcing feature-space alignment at several hierarchical levels. On the other hand, face_loss5 exhibits slightly more noise in background areas (clothing patterns, hair texture), which is in line with the -4.14% SSIM degradation on Medium.

Figure 5.7: Consistent gains from face_loss5 are shown by side-by-side comparisons across multiple identities. Zooming into insets (bottom row) reveals crisper eyelid folds and eyebrow texture, which are traits known to carry strong identity signals

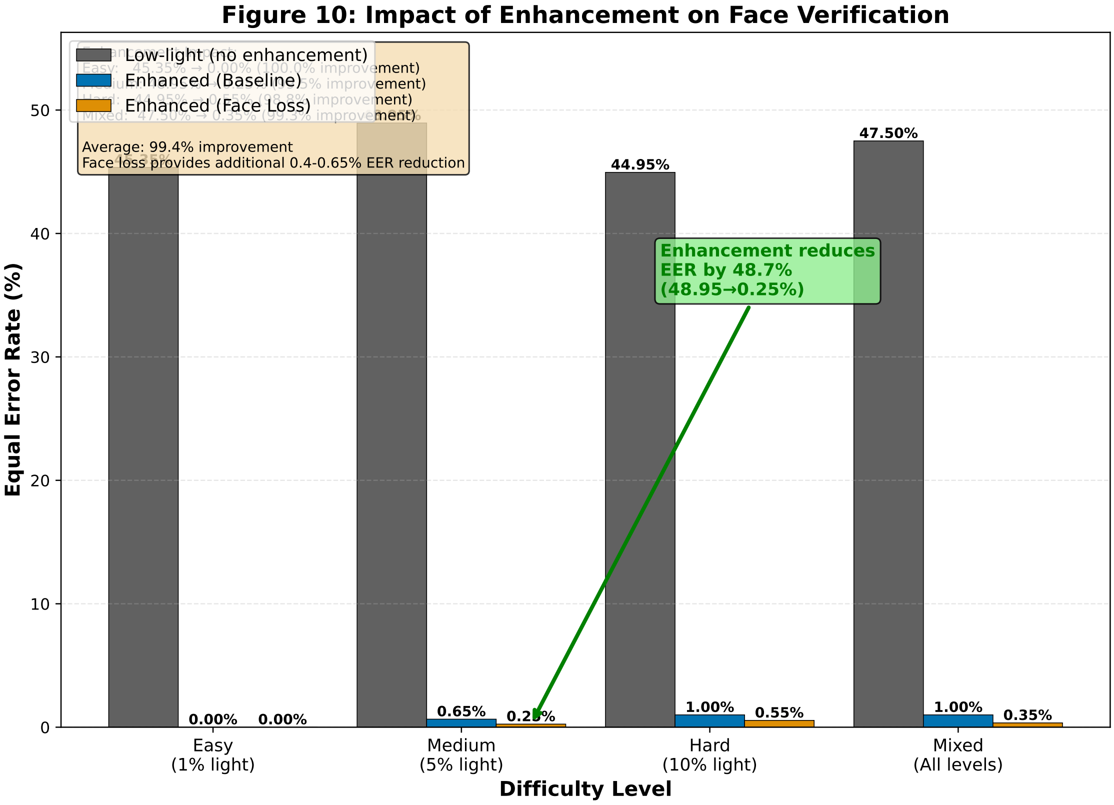

Figure 5.6: Visual comparison of baseline vs. face_loss5 enhancement on Medium difficulty.

[42]. Although imperceptible to human observers, these perceptual variations result in quantifiably higher similarity scores for same-identity pairs (0.9675 vs. 0.9519 on Medium), hence improving verification accuracy.

Figure 5.8: The gallery illustrates the constant nature of Face_loss5 across different face traits (race, age, gender, and eyewear). Within-row similarity (many images of the same person) verifies strong identity preservation, and between-row variation validates generalization to various identities. The infrequent artifacts that appear in severe shadows (behind the chin, hair borders) reflect the basic information-theoretic limit, which states that areas with nearly zero photon counts in the input produce minimal signal for reconstruction, regardless of model sophistication.

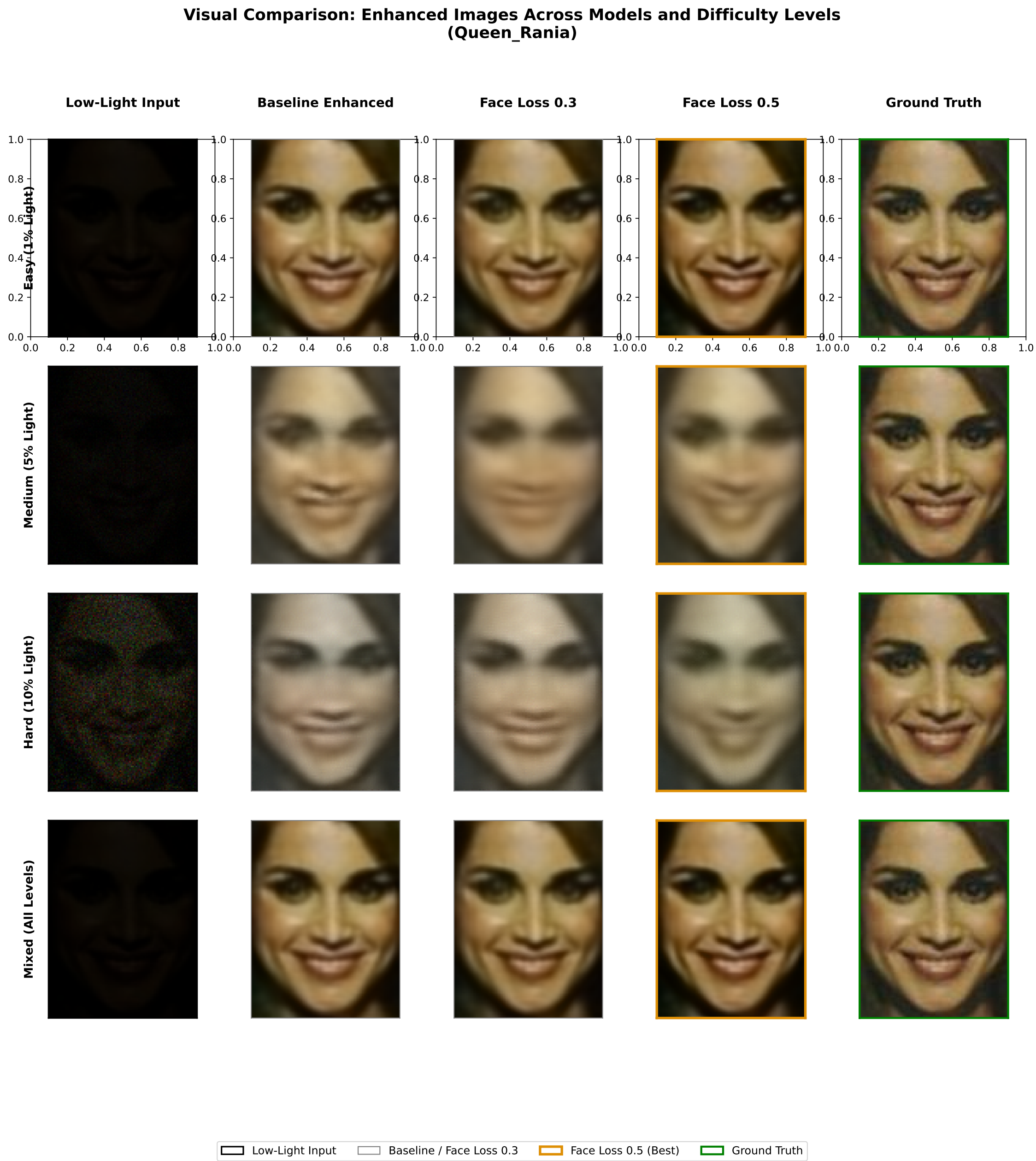

FIGURE 5.7: Side-by-side comparison of input low-light images, baseline enhancements, and face_loss5 enhancements across multiple identities and difficulty levels.

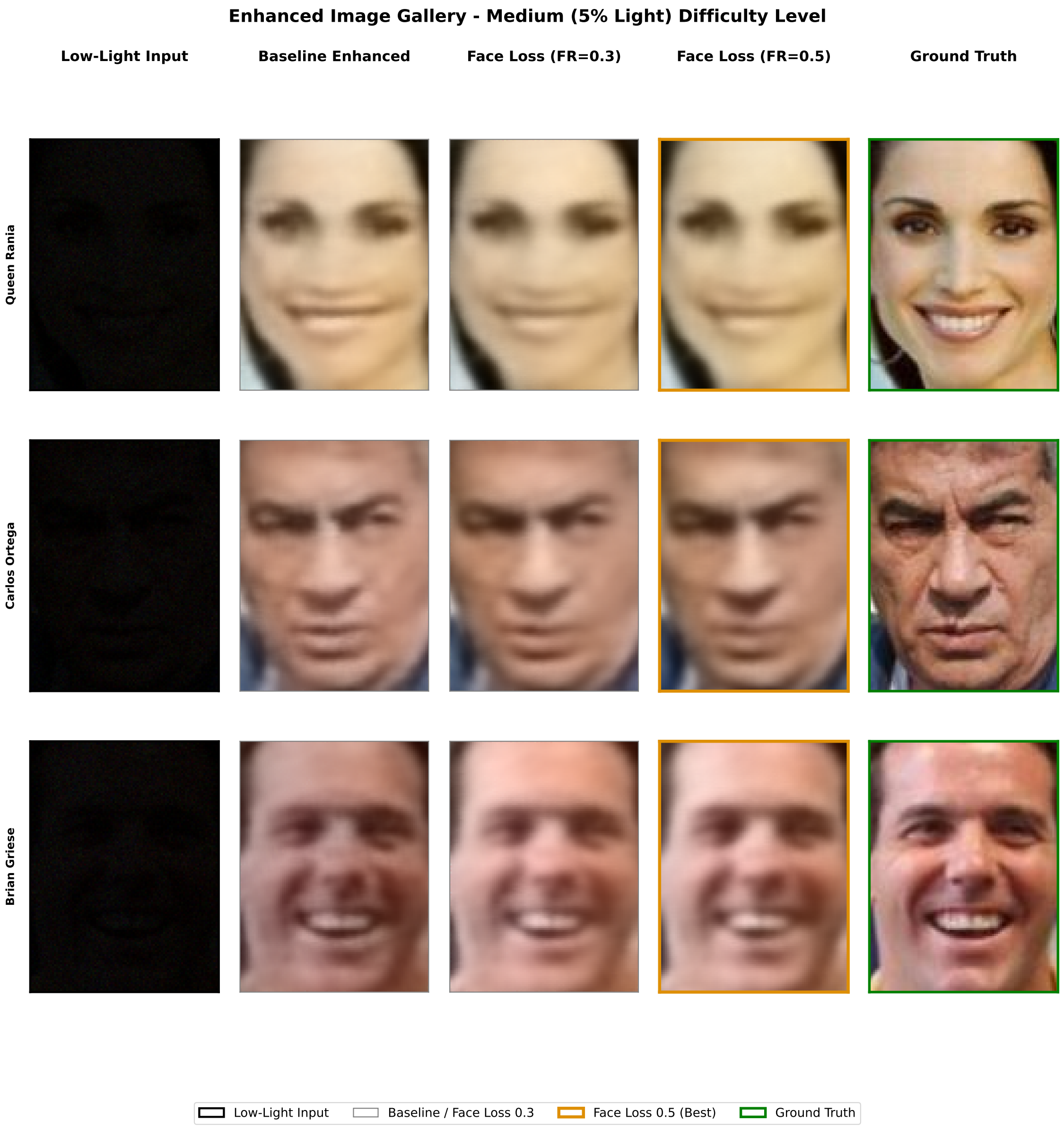

FIGURE 5.8: Gallery of face_loss5 enhancements on Medium difficulty across diverse identities, poses, and expressions.

Chapter 6

Conclusions

6.1 Research Summary

This study tackles the fundamental issue of maintaining facial identity information during low-light picture enhancement, which is crucial for face recognition systems functioning under difficult lighting circumstances. Our research shows that typical enhancement losses compress the feature space of facial embeddings, decreasing inter-class separability and verification accuracy, in contrast to standard enhancement algorithms that optimize just for perceptual quality. In order to overcome this constraint, we developed and put into practice a discriminative multi-level face loss function that uses facial features taken from several layers of a frozen face recognition network to actively oversee the augmentation process.

Our fundamental premise that discriminative facial traits are preserved throughout enhancement by explicit identity monitoring is supported by these quantitative findings. From low-level texture (layer 2) to high-level identity representations (fc layer), the multi-level feature reconstruction loss guarantees that improved images retain semantic correspondence with ground truth at all network depths. By drawing real pairs together and pushing fake pairs apart in embedding space, the supervised contrastive loss prevents the feature compression seen in traditional enhancement and enforces inter-class separation. In order to preserve discriminative capability even in the face of extreme degradation, the triplet margin loss adds another geometric constraint by requiring a minimum distance between positive and negative pairings.

According to the trade-off study, higher face loss weights ($\omega_{FR} = 0.5$) result in significant gains in verification accuracy but a minor decline in perceptual quality measures (SSIM drops from 0.8262 to 0.8048 on mixed difficulty). For security-critical applications like surveillance systems, access control, and law enforcement, where the accuracy of identity verification is more important than subjective visual quality, this trade-off is acceptable—even desirable. The slight increases in PSNR (27.41 dB to 27.88 dB) show that discriminative face loss greatly improves recognition ability without substantially sacrificing picture reconstruction quality.

This study shows that task-specific supervision greatly outperforms generic perceptual quality optimization for downstream recognition applications. Future additions to different face recognition architectures, alternative enhancement networks, and video-based temporal consistency techniques are made possible by the modular framework design, which offers a sensible method of maintaining identification information under difficult imaging circumstances.

6.2 Contributions and Impact

The key contributions of this research are:

Discriminative Multi-Level Face Loss: We propose a combination of triplet loss, supervised contrastive loss and reconstruction loss to generate a unique loss function that explicitly retains face identification information throughout enhancement.

Physics-Based Low-Light Synthesis: We create a physics-based low-light image synthesis pipeline.

Empirical Validation and Quantitative Results: We validate the effectiveness of discriminative supervision for identity preservation.

This work has wider applications for biometric systems operating in difficult environments, such as autonomous systems needing reliable person identification, nighttime surveillance, indoor access control with dim lighting, and forensic image analysis.

6.3 Limitations and Challenges

Despite of the improvements, the limitations and challenges faced are:

Computational Overhead: Current architecture is not suitable for real-time challenges due to high cost of multilevel feature extraction.

Single Dataset Evaluation: Lack of real-life low-light face dataset steered us to proceed with synthetic dataset generation. Evaluation on independently captured low-light face images would strengthen claims of real-world applicability.

6.4 Future Research Directions

Future studies could focus on:

Temporal Consistency for Video Enhancement: Using optical flow-based temporal alignment to handle video sequences

Cross-Architecture Generalization: Evaluating the discriminative face loss framework using face recognition models other than AdaFace.

Adaptive Face Loss Weighting: Developing adaptive systems that automatically adjust ω_{FR} in response to measures of image quality or the degree of deterioration

Real-World Dataset Validation: Assessing generality beyond synthetic degradations, a comprehensive analysis is conducted on independently gathered low-light face datasets.

Acknowledgement of AI Usages

Generative AI tools (e.g., ChatGPT, Gemini) were used to (i) correct grammatical errors, (ii) assist in restructuring paragraphs to improve academic readability, and (iii) paraphrase selected passages to enhance clarity and conciseness. All AI-generated suggestions were carefully reviewed, edited and validated by the authors prior to inclusion. No new technical content, claims, or conclusions were generated by the AI tools.

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